

Original research article

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ABSTRACT

This work examines a nonlinear stochastic SIRS epidemic model evolving in a randomly changing environment described by a finite-state Markov chain. The transmission mechanism incorporates regime-dependent nonlinear incidence rates, where the contact interaction between susceptible and infectious individuals is modeled by the term $S^{\rho_{\xi(t)}} I^{\zeta_{\xi(t)}}$. This switching nonlinearity allows the model to capture varying environmental effects and heterogeneous transmission patterns more accurately, thereby providing a more realistic description of epidemic dynamics. To the best of our knowledge, the sufficient criteria governing the persistence and extinction of stochastic SIRS models with transmission rate exponents governed by Markovian switching have not yet been established in the existing literature. The principal contribution of the present study is the derivation of the rigorous sufficient conditions that characterize both the extinction and the long-term persistence of disease dynamics. Specifically, a threshold parameter Λ , expressed in terms of the switching exponents $\rho_{\xi(t)}$ and $\zeta_{\xi(t)}$, is derived. That is, if $\Lambda > 0$, the disease exhibits strong stochastic persistence; conversely, if $\Lambda < 0$, the disease-free equilibrium state becomes globally asymptotically stable in probability, leading to eventual disease extinction. In the special case where there is no regime switching and $\rho_{\xi(t)} = \zeta_{\xi(t)} = 1$, our model recovers the classical threshold found in the literature. To support and validate the theoretical findings, numerical simulations are provided to demonstrate the dynamical behavior of the model under different environmental regimes.

1. Introduction

Infectious diseases have long constituted major challenges to global public health and human society. The analysis and control of these diseases require an interdisciplinary approach, in which mathematical modeling has emerged as an indispensable tool for investigating epidemic spread and transmission dynamics [1,2]. Mathematical epidemic models offer a rigorous and quantitative framework for describing the mechanisms governing disease propagation within populations, assessing the evolution of outbreaks, and predicting their potential impact. Consequently, these models provide valuable insights that support evidence-based decision-making and the design of effective prevention and control strategies.

In 1760, Daniel Bernoulli introduced one of the earliest mathematical approaches to epidemiology through his study of smallpox transmission and the impact of inoculation. This pioneering contribution laid the foundation for modern epidemic modeling. A major advancement in this field was later achieved by William O. Kermack and Anderson

G. McKendrick in 1927 with the formulation of the classical SIR compartmental model [3]. In this framework, the population is divided into three epidemiological classes: susceptible (S), infected (I), and recovered or removed (R). The model assumes that susceptible individuals acquire the infection through contact with infectious individuals, while infected individuals eventually recover with permanent immunity or are removed from the population. Owing to its conceptual simplicity and strong analytical foundation, the SIR framework has become a cornerstone of mathematical epidemiology. It has inspired numerous extensions and applications in the study of infectious disease dynamics [4]. Although the traditional SIR model (Susceptible-Infected-Recovered) assumes permanent immunity after recovery, real-world evidence indicates that immunity to certain diseases, such as influenza [5], viral diarrheal infections [6], and foot and mouth disease [7], can be temporary. In these instances, recovered individuals can eventually lose immunity and rejoin the susceptible population, prompting the development of the SIRS (Susceptible-Infected-Recovered-Susceptible) model to better capture these dynamics [8]. The incidence rate plays an essential role

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in understanding the spread of infectious diseases and has been the subject of significant research. In particular, nonlinear incidence rates have offered a more realistic representation of epidemic dynamics. In this context, the SIRS model with nonlinear incidence functions reveals a rich spectrum of dynamic behaviors, including bifurcations, oscillations, and multiple equilibria. This framework improves our understanding of epidemic dynamics by capturing the complexities of disease spread under varying assumptions. Future work should explore the implications of these dynamics for real-world disease control and prevention strategies. The stochastic perturbation of the incidence exponents, i.e. the shift from $S^\rho I^\zeta$ to $S^{\rho_{\xi(t)}} I^{\zeta_{\xi(t)}}$, does not reflect a direct individual-level biological mechanism. Rather, it models temporal fluctuations in transmission due to environmental and behavioural heterogeneity—encompassing seasonality, contact modifications, public health measures, and pathogen or host variability across distinct contexts. The switching process $\xi(t)$ denotes transitions between epidemiological or environmental states, each associated with specific transmission exponents (ρ_j, ζ_j) . Such a regime-switching formulation is standard for representing abrupt or discrete alterations in the transmission setting, as opposed to intrinsic microscopic changes in the pathogen or host. With regard to disease specificity, the proposed framework is conceived as a broad stochastic epidemic model, not confined to any particular infection. Nevertheless, it proves especially relevant for pathogens whose transmission is heavily modulated by environmental or social factors—for example, influenza-like illnesses, vector-borne diseases, or HIV/AIDS under varying behavioural and intervention scenarios. Although standard incidence rates of the form βSI have received attention in works such as [9–11]. In addition, alternative nonlinear formulations, including $\beta S^\rho I^\zeta$, have been introduced to better represent the intricacies of disease transmission dynamics [12–14]. In this work, we study a specific nonlinear incidence of the form $\beta S^\rho I^\zeta$, which is a particular case of a general incidence function $f(S, I)$, but with Markovian switching. The general function $f(S, I)$ denotes the force of infection and is assumed to satisfy standard conditions such as positivity and sufficient regularity to ensure biological consistency. By contrast, the model examined in this paper adopts the explicit nonlinear form $\beta S^\rho I^\zeta$, where $\beta > 0$, $\rho > 0$, and $\zeta > 0$. This functional form serves as a special instance of the general incidence framework and is chosen to capture nonlinear effects in disease transmission, including heterogeneous contact patterns and saturation phenomena. Hence, while $f(S, I)$ outlines the general modeling structure, all analytical results derived in this paper are established specifically for the incidence $\beta S^\rho I^\zeta$. For a broader perspective on general incidence functions, foundational contributions can be found in [13,15–18]. The nonlinear incidence function $\beta S^\rho I^\zeta$ was investigated by Liu et al. [19,20] within the framework of the following system of differential equations

$$\begin{cases} \frac{dS}{dt} = \mu - \mu S - \beta S^\rho I^\zeta + \eta R, \\ \frac{dI}{dt} = -(\mu + \delta)I + \beta S^\rho I^\zeta, \\ \frac{dR}{dt} = -(\mu + \eta)R + \delta I, \end{cases}$$

where, μ represents the natural death rate of the population (per capita death rate), β denotes the infection transmission rate, indicating how quickly susceptible individuals become infected through interaction with infected individuals, ρ describes the effect of the susceptible population S on the transmission dynamics, often referred to as the susceptibility exponent. Similarly, ζ represents the effect of the infected population I on the transmission dynamics, often referred to as the infectivity exponent, η is the recovery rate from the recovered compartment R to the susceptible compartment S , representing loss of immunity or waning immunity. Finally, δ is the recovery rate from the infected compartment I to the recovered compartment R , representing individuals recovering from the disease and gaining immunity. The authors highlight significant changes in solution behavior as the incidence rate transitions from the classical bilinear mass action form, βIS , to a generalized nonlinear rate, $\beta I^\zeta S^\rho$. The primary lesson of Liu et al. [19,20] is that the behavior

of the solutions is governed by the following quantities.

$$\begin{aligned} \mathcal{T}^* &= \frac{(\zeta + \rho - 1)^{\zeta + \rho - 1}}{(H(\zeta - 1))^{\zeta - 1} \rho^\rho}, \\ \mathcal{T} &= \frac{\beta}{\mu + \delta}, \\ H &= \frac{\mu + \eta}{\delta + \mu + \eta}. \end{aligned}$$

For $\zeta = 1$, the disease dies if $\mathcal{T} \leq 1$, but remains endemic and approaches a unique equilibrium if $\mathcal{T} > 1$. For $0 < \zeta < 1$, the threshold condition completely disappears, as the disease always remains endemic and becomes closer to a special equilibrium. The threshold result becomes more complicated when $\zeta > 1$. In particular:

- The disease dies if $\mathcal{T} < \mathcal{T}^*$.
- When $\mathcal{T} > \mathcal{T}^*$ the disease dies for some initial conditions, but remains endemic and approaches the larger non-trivial equilibrium. The smaller nontrivial equilibrium is a saddle, whereas the larger nontrivial equilibrium may be locally asymptotically stable in some cases and unstable in others.
- When $\mathcal{T} = \mathcal{T}^*$, a saddle-node bifurcation occurs.
- When $\mathcal{T} \leq 1$, the trivial equilibrium is locally asymptotically stable and becomes an unstable saddle as ζ decreases through 1, while the nontrivial equilibrium is locally asymptotically stable
- When $\mathcal{T} > 1$, the trivial equilibrium is an unstable saddle for $\zeta < 1$, and it becomes locally asymptotically stable as ζ increases through 1, while an unstable nontrivial equilibrium bifurcates from it.

In the study of infectious disease dynamics, the parameters of the SIRS model are not constant but are influenced by environmental variability. Stochastic approaches have been developed to account for these fluctuations, including random perturbations. In their work [21], the authors built on the traditional SIRS model by incorporating stochastic perturbations into the infection rate. Specifically, the infection coefficient β is replaced by $\beta + \sigma \frac{dW}{dt}$, where W denotes a Brownian motion and σ represents the intensity of the noise, capturing the extent of random variability within the system. In other words, it quantifies how strongly the evolution of β is influenced by stochastic fluctuations. It also reflects the inherent uncertainty in the transmission dynamics. This perturbation leads to the following system.

$$\begin{cases} dS_t = [\mu - \mu S_t - \beta S_t I_t + \eta R_t]dt - \sigma S_t I_t dW_t, \\ dI_t = [-(\mu + \delta)I_t + \beta S_t I_t]dt + \sigma S_t I_t dW_t, \\ dR_t = [-(\mu + \eta)R_t + \delta I_t]dt. \end{cases} \quad (1)$$

Numerous researchers have studied the SIRS model with Brownian perturbations, investigating its stability and dynamic behaviors. In [22–25], the authors analyzed various aspects of the model, including its equilibrium states and the impact of stochastic effects on the dynamics of the epidemic. Building on these foundational studies, the SIRS model (1) was further expanded by Lahrouz et al. [26], who introduced a saturated incidence rate and disease-related mortality. They also established global stability conditions for the equilibrium state E_0 , both in probability and at the q -th moment. In [27], Lahrouz and Settati refined the earlier findings in the literature by deriving the necessary condition that governs the dichotomy between the extinction and persistence of the epidemic. Furthermore, they formulated an exact expression for the critical thresholds that determine the asymptotic dynamics of the SIRS system, represented by the following stochastic threshold.

$$\mathcal{R}_s \triangleq \frac{\beta}{\mu + \delta + \frac{1}{2}\sigma^2}.$$

Beyond white noise, environmental fluctuations can be modeled using colored noise, such as telegraph noise [28–30]. This type of noise reflects the alternations between different environmental regimes, which

can correspond to variations in factors such as nutrition, climate, or sociocultural conditions. These transitions are typically memoryless, with waiting times between regime changes following an exponential distribution, thereby introducing additional complexity into the modeling of infectious disease dynamics. Recent studies have significantly improved our understanding of the SIRS model by incorporating nonlinear incidence rates. Notable contributions include those by Cui and Zhao [31], Wang et al. [32], Liu et al. [33], Pan et al. [34], and Wang et al. [35]. Recently, several studies have contributed to the advancement of mathematical modeling of cholera and related epidemiological phenomena. In particular, the works explore different modeling approaches, including fractional calculus and nonlinear dynamics, to better capture the complexities of disease transmission and control. However, to our knowledge, none of these studies has considered a non-linear binomial-type incidence rate, where the exponents of susceptible and infected populations are strictly positive and coupled through a Markov chain. To address this gap, we propose an innovative formulation for the incidence rate that incorporates a generalized nonlinear rate, expressed as $S^{\rho_{\xi(t)}} I^{\zeta_{\xi(t)}}$, where the exponents are strictly positive and depend on the state of the Markov chain. We analyze the asymptotic behavior of the system under stochastic perturbations and establish the sufficient conditions for disease extinction and persistence. This approach provides valuable information on the long-term effects of environmental fluctuations on epidemic outcomes. Telegraph noise, characterized by abrupt transitions between environmental states, influences disease transmission rates, either amplifying or mitigating the spread. These regime shifts are modeled as transitions in a finite-state Markov chain. To begin with, let $\xi(t)$ denote a Markov chain defined in a finite state space $\mathcal{E} = \{1, 2, \dots, r\}$ with the generator matrix $\Gamma = (\gamma_{ij})_{1 \leq i, j \leq r}$. The transition probabilities over a short time interval $\varsigma > 0$ are given by

$$\begin{aligned} \mathbb{P}(\xi(t + \varsigma) = j \mid \xi(t) = i) \\ = \begin{cases} \gamma_{ij}\varsigma + o(\varsigma), & \text{if } i \neq j, \\ 1 + \gamma_{ii}\varsigma + o(\varsigma), & \text{if } i = j, \end{cases} \quad \text{where } \gamma_{ii} = -\sum_{j \neq i} \gamma_{ij}. \end{aligned} \quad (2)$$

Assuming the irreducibility of the Markov chain $\xi(t)$, the system admits a unique stationary distribution $\pi = (\pi_1, \pi_2, \dots, \pi_r) \in \mathbb{R}^r$. The irreducibility of the Markov chain $\xi(t)$ is a standard and realistic assumption in this framework, as it reflects the ability of the environmental system to switch recurrently between its different regimes. This property is fundamental for ensuring the existence of a unique stationary distribution π , which in turn makes it possible to define the global threshold Λ . Moreover, irreducibility allows the application of Birkhoff's ergodic theorem, a key ingredient in the proofs of Theorems 2, 3, and 5. Consequently, we present a stochastic SIRS model characterized by a nonlinear incidence rate, with its parameters dynamically governed by the state of the Markov chain $\xi(t)$. The dynamics of the model are characterized by the following system of differential equations.

$$\begin{cases} dS_t = \left(\mu_{\xi(t)} - \mu_{\xi(t)} S_t - \beta_{\xi(t)} S_t^{\rho_{\xi(t)}} I_t^{\zeta_{\xi(t)}} + \eta_{\xi(t)} R_t \right) dt - \sigma_{\xi(t)} S_t^{\rho_{\xi(t)}} I_t^{\zeta_{\xi(t)}} dW_t, \\ dI_t = \left(-(\mu_{\xi(t)} + \delta_{\xi(t)}) I_t + \beta_{\xi(t)} S_t^{\rho_{\xi(t)}} I_t^{\zeta_{\xi(t)}} \right) dt + \sigma_{\xi(t)} S_t^{\rho_{\xi(t)}} I_t^{\zeta_{\xi(t)}} dW_t, \\ dR_t = \left(-(\mu_{\xi(t)} + \eta_{\xi(t)}) R_t + \delta_{\xi(t)} I_t \right) dt. \end{cases} \quad (3)$$

For each state $i \in \mathcal{E}$, the parameters $\mu_i, \beta_i, \rho_i, \zeta_i, \eta_i, \delta_i$ and σ_i represent positive constants, retaining the same meanings as previously defined. Initially, at time τ_0 , if the Markov chain 4, 5, 6, is in state $\xi(\tau_0) = i_0$ and transits, in time τ_k , to a new state $\xi(\tau_k) = j$, where $j \in \mathcal{E}$, the system (3) is well described by the following stochastic differential equations under regime switching (Figs. 1, 4–14, and 17).

$$\begin{cases} dS_t = \left(\mu_j - \mu_j S_t - \beta_j S_t^{\rho_j} I_t^{\zeta_j} + \eta_j R_t \right) dt - \sigma_j S_t^{\rho_j} I_t^{\zeta_j} dW_t, \\ dI_t = \left(-(\mu_j + \delta_j) I_t + \beta_j S_t^{\rho_j} I_t^{\zeta_j} \right) dt + \sigma_j S_t^{\rho_j} I_t^{\zeta_j} dW_t, \\ dR_t = \left(-(\mu_j + \eta_j) R_t + \delta_j I_t \right) dt. \end{cases} \quad (4)$$

The primary challenge in analyzing such a stochastic model lies in determining the conditions under which the epidemic either dies out or persists within the population. Until now, the sufficient conditions for the persistence and extinction of the system (3) have not been explored in the literature. The primary objective of this study is to rigorously establish these sufficient conditions for the stochastic SIRS epidemic model (3), which incorporates transmission rate exponents governed by Markovian switching dynamics. Our work presents the first comprehensive and rigorous demonstration of these conditions, addressing a significant gap in understanding the mechanisms of extinction and persistence in such systems. This work is organized as follows in Section 2, we present the problem formulation and outline the auxiliary results that serve as the foundation for the subsequent analysis. Section 3 explores the conditions under which the epidemic model (3) becomes extinct, employing various convergence criteria. In Section 4, we establish the criteria required for the persistence of the epidemic. Numerical simulations are provided in Section 5 to illustrate and validate the theoretical findings. Finally, Section 6 concludes with a summary of the main results and a discussion of their broader implications.

2. Formulation and auxiliary results

To begin with, we denote $\mathbb{R}_+^3 = \{x \mid x_i > 0, i = 1, 2, 3\}$ and $\Delta = \{x \in \mathbb{R}_+^3 \mid x_1 + x_2 + x_3 = 1\}$. Let $X(t)$ represent a regular time-homogeneous Markov process in $\mathbb{R}_+^3$, which can be characterized by the following stochastic differential equations.

$$dX(t) = b(X(t), \xi(t)) dt + h(X(t), \xi(t)) dW(t), \quad (5)$$

where $X(t) = (S(t), I(t), R(t))$. Here, $W(t)$ is a standard Brownian motion defined in the complete probability space $(\Omega, \mathcal{F}, (F_t)_{t \geq 0}, \mathbb{P})$, and $(\xi(t))_{t \geq 0}$ is a right-continuous Markov chain in the same probability space, taking values in the finite state space $\mathcal{E} = \{1, 2, \dots, r\}$. The Markov chain is characterized by the generator matrix $\Gamma = (\gamma_{ij})_{1 \leq i, j \leq r}$, as defined in Eqs. (2). Following the framework introduced by Zhu and Yin [36], the diffusion matrix $A(x, j)$ for each $j \in \mathcal{E}$ and $x \in \mathbb{R}_+^3$ is given by

$$A(x, j) = h(x, j)h^T(x, j).$$

For the purpose of the subsequent analysis, we introduce the generator $\mathcal{L}$ associated with the SDE system in Eq. (5). For each $j \in \mathcal{E}$, $x \in \mathbb{R}_+^3$ and any twice continuously differentiable function $V(x, j)$, the generator $\mathcal{L}$ is defined as

$$\begin{aligned} \mathcal{L}V(x, j) &= b^T(x, j) \cdot \nabla V(x, j) + \frac{1}{2} \text{Tr}(h^T(x, j) \cdot \nabla^2 V(x, j) \cdot h(x, j)) \\ &\quad + \Gamma(x)V(x, \cdot)(j), \end{aligned}$$

where

$$\Gamma(x)V(x, \cdot)(j) = \sum_{k \neq j, k \in \mathcal{E}} \gamma_{jk}(V(x, k) - V(x, j)).$$

To ensure the biological relevance of model (3), it is necessary to prove that every solution remains confined to Δ . Following Khasminskii et al. [37] and Yuan and Mao [38], we state the following conditional result.

Theorem 1. *Provided that the positive invariance of Δ holds (see Remark 1), for any initial condition $(S_0, I_0, R_0, \xi_0) \in \Delta \times \mathcal{E}$, we have*

$$\mathbb{P}((S(t), I(t), R(t), \xi(t)) \in \Delta \times \mathcal{E}, \forall t \geq 0) = 1.$$

Remark 1. The rigorous proof is nontrivial when ρ_j or $\zeta_j < 1$, since the coefficients $S^{\rho_j} I^{\zeta_j}$ are not locally Lipschitz in the neighborhood of $S = 0$ or $I = 0$. The classical Khasminskii argument does not cover all parameter ranges. The subsequent use of $\log I$ in (52), I^φ in Theorem 4, and $I^{-\alpha}$ in (57)–(60) requires that the solution does not hit the boundary. A complete proof, via a Lyapunov function or a localization procedure, is deferred to future work. In this paper, we proceed under this hypothesis, which is standard in the literature and supported by simulations.

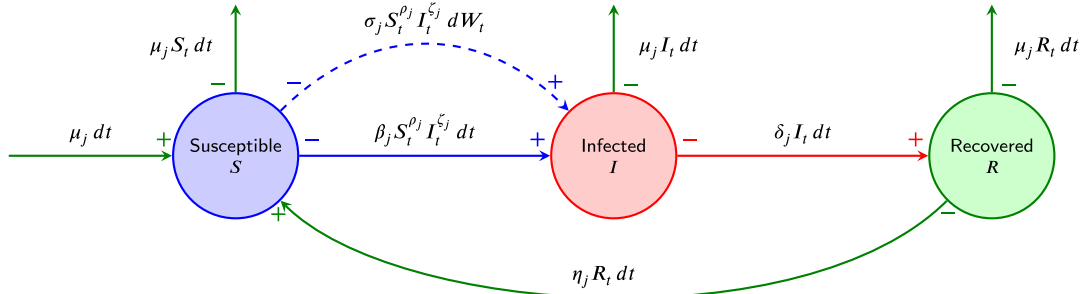


Fig. 1. Diagram of the SIRS model with a nonlinear exponent in the transmission rate.

Let us introduce the following lemma, which will be instrumental in proving the extinction condition. To begin with, let us consider, for $p, q > 0$, the function $f_{(p,q)}$ defined by

$$f_{(p,q)}(x, y) = x^p y^q, \quad \text{for } x, y \in D,$$

where

$$D = \{(x, y) \in \mathbb{R}_+^2; x + y \leq 1\}.$$

Lemma 1. Let $p, q > 0$. The function $f_{(p,q)}$, defined in the domain D , attains its global maximum at the critical point $(\frac{p}{p+q}, \frac{q}{p+q})$. The maximum value of $f_{(p,q)}$ at this critical point is given by

$$f_{(p,q)}\left(\frac{p}{p+q}, \frac{q}{p+q}\right) = \frac{p^p q^q}{(p+q)^{p+q}}.$$

Moreover, the function $f_{(p,q)}$ attains the global minimum 0 when $x = 0$ or $y = 0$, and we have the following inequalities for $x \in (0, \frac{p}{p+q}]$ and $y \in (0, \frac{q}{p+q}]$,

$$f_{(p,q)}(x, y) \leq f_{(p,q)}\left(x, \frac{q}{p+q}\right) \leq f_{(p,q)}\left(\frac{p}{p+q}, \frac{q}{p+q}\right),$$

$$f_{(p,q)}(x, y) \leq f_{(p,q)}\left(\frac{p}{p+q}, y\right) \leq f_{(p,q)}\left(\frac{p}{p+q}, \frac{q}{p+q}\right).$$

Proof. Consider the set $D = \{(x, y) \in \mathbb{R}_+^2 \mid x + y \leq 1\}$. The closure of D , denoted $\bar{D}$, is both closed and bounded in $\mathbb{R}^2$, implying that $\bar{D}$ is compact by the Borel theorem. The function $f_{(p,q)}(x, y) = x^p y^q$ is continuous on $\bar{D}$. Consequently, $f_{(p,q)}$ is bounded on $\bar{D}$ and attains its extrema in this compact domain. On the boundary of D , where $x = 0$ or $y = 0$, we observe that $f_{(p,q)}(x, y) = 0$. Moreover, since the partial functions $f_{(p,q)}(x, \cdot)$ and $f_{(p,q)}(\cdot, y)$ are increasing with respect to x and y . Consequently, the maximum of $f_{(p,q)}$ must occur along the boundary of the line $x + y = 1$ within the domain D . To determine the critical points of $f_{(p,q)}$, we compute its partial derivatives. Differentiating with respect to x , we compute:

$$\begin{aligned} \frac{df_{(p,q)}(x, y)}{dx} &= \frac{\partial f_{(p,q)}(x, y)}{\partial x} + \frac{\partial y}{\partial x} \frac{\partial f_{(p,q)}(x, y)}{\partial y} \\ &= px^{p-1}y^q - qx^p y^{q-1}. \end{aligned}$$

Setting this derivative equal to zero and using the constraint $x + y = 1$, we obtain

$$x = \frac{p}{p+q} \quad \text{and} \quad y = \frac{q}{p+q}.$$

Therefore, the critical point is $(\frac{p}{p+q}, \frac{q}{p+q})$. Next, we evaluate $f_{(p,q)}$ along the boundary of D , where $x + y = 1$. At the critical point, we have

$$f_{(p,q)}\left(\frac{p}{p+q}, \frac{q}{p+q}\right) = \frac{p^p q^q}{(p+q)^{p+q}}.$$

It is clear that as $x \rightarrow 0$ or $y \rightarrow 0$, the function tends to zero, that is, $f_{(p,q)}(0, y) = 0$ or $f_{(p,q)}(x, 0) = 0$. By the Extreme Value Theorem, since $f_{(p,q)}$ is continuous on $\bar{D}$, it attains a global maximum and a global minimum in $\bar{D}$. Therefore, the global maximum $\frac{p^p q^q}{(p+q)^{p+q}}$ occurs at the point

$(\frac{p}{p+q}, \frac{q}{p+q})$, and the global minimum 0 is attained at the boundary of D . $\square$

To analyze the stability properties of the disease-free equilibrium, we invoke the following result based on Lyapunov functions.

Lemma 2. If there exist a Lyapunov function $V \in C^2(\mathbb{R}^3 \times \mathcal{E}; \mathbb{R}^+)$ and a function $w : \mathbb{R}^3 \rightarrow \mathbb{R}^+$ that is zero only at the equilibrium point E_0 . If the following conditions are satisfied

$$\mathcal{L}V(x, j) \leq -w(x), \quad \text{for all } (x, j) \in \mathbb{R}^3 \times \mathcal{E}, \quad \text{and} \quad \lim_{|x| \rightarrow \infty} \inf_{j \in \mathcal{E}} V(x, j) = \infty,$$

then, the equilibrium state E_0 of the system (3) is globally asymptotically stable in probability.

This lemma provides a sufficient condition for global asymptotic stability in probability, a concept rigorously formulated by Khasminskii et al. [37]. Next, we derive the threshold condition, denoted by Λ , between extinction and persistence of the disease, expressed in terms of the model parameters as:

$$\Lambda = \begin{cases} \sum_{j=1}^r \pi_j \tilde{\Lambda}_j, & \text{if } \zeta_j < 1, \\ \sum_{j=1}^r \pi_j \bar{\Lambda}_j, & \text{if } \zeta_j \geq 1, \end{cases} \quad (6)$$

where

$$\tilde{\Lambda}_j = \frac{\beta_j^2}{2(1 - \zeta_j)\sigma_j^2} - (\mu_j + \delta_j), \quad (7)$$

$$\bar{\Lambda}_j = \begin{cases} \frac{\beta_j^2}{2\sigma_j^2} - (\mu_j + \delta_j), & \text{if } \frac{\beta_j}{\sigma_j^2} \leq \frac{\rho_j^{\rho_j} (\zeta_j - 1)^{\zeta_j - 1}}{(\rho_j + \zeta_j - 1)^{\rho_j + \zeta_j - 1}}, \\ \Theta_j \left(\frac{\rho_j^{\rho_j} (\zeta_j - 1)^{\zeta_j - 1}}{(\rho_j + \zeta_j - 1)^{\rho_j + \zeta_j - 1}} \right), & \text{if } \frac{\beta_j}{\sigma_j^2} > \frac{\rho_j^{\rho_j} (\zeta_j - 1)^{\zeta_j - 1}}{(\rho_j + \zeta_j - 1)^{\rho_j + \zeta_j - 1}}, \end{cases} \quad (8)$$

and

$$\Theta_j(x) = -\frac{1}{2}\sigma_j^2 x^2 + \beta_j x - (\mu_j + \delta_j), \quad \forall x \in \mathbb{R}, \quad j \in \mathcal{E}.$$

Remark 2. It is important to emphasize that the threshold Λ does not depend directly on the comparison between ρ_j and ζ_j for a fixed regime j . Indeed, when $\zeta_j < 1$, the local threshold $\tilde{\Lambda}_j$ is independent of ρ_j . When $\zeta_j \geq 1$, the dependence on ρ_j appears through the term

$$K_j = \frac{\rho_j^{\rho_j} (\zeta_j - 1)^{\zeta_j - 1}}{(\rho_j + \zeta_j - 1)^{\rho_j + \zeta_j - 1}},$$

which represents the maximum of $S^{\rho_j} I^{\zeta_j - 1}$ over the domain $S + I \leq 1$. The relative order $\rho_j < \zeta_j$ or $\rho_j \geq \zeta_j$ influences the location of this maximum, and consequently the value of K_j , thereby modifying the threshold and the resulting extinction or persistence conditions 3. However, no case distinction based solely on ρ_j relative to ζ_j is required in the definition of Λ .

3. Extinction

In this section, we analyze the extinction condition 5 of system (3). To begin with, let us cite the following theorem investigating the global stability of the disease-free equilibrium 7, 8, 9, 10, 11.

3.1. Global stability of the disease-free equilibrium state

Theorem 2. Let $(S_0, I_0, R_0, \xi_0) \in \Delta \times \mathcal{E}$. The disease-free equilibrium state E_0 is globally asymptotically stable in probability if one of the following conditions is satisfied

1. If $\zeta_j < 1$, for all $j \in \mathcal{E}$, and

$$\Lambda = \sum_{j=1}^r \pi_j \tilde{\Lambda}_j < 0. \quad (9)$$

2. If $\zeta_j \geq 1$, for all $j \in \mathcal{E}$, and

$$\Lambda = \sum_{j=1}^r \pi_j \bar{\Lambda}_j < 0. \quad (10)$$

Proof of Theorem 2. Let $(S_0, I_0, R_0) \in \Delta$. We define the following Lyapunov function

$$V(S, I, R, j) = \vartheta_1(1 - S)^2 + \left(\frac{1}{\varphi} + b_j\right) I^\varphi + \vartheta_2 R^2, \quad \forall j \in \mathcal{E},$$

where $\vartheta_1 > 0$, $\varphi > 0$, $\vartheta_2 > 0$ and b_j are real values that will be identified in the subsequent steps. From Itô's formula, we have

$$\begin{aligned} \mathcal{L}V = & -2\vartheta_1\mu_j(1 - S)^2 + 2\vartheta_1\beta_j S^{\varrho_j} I^{\zeta_j} (1 - S) - 2\vartheta_1\eta_j R(1 - S) + \vartheta_1\sigma_j^2 S^{2\varrho_j} I^{2\zeta_j} \\ & - \varphi \left(\frac{1}{\varphi} + b_j\right) (\mu_j + \delta_j) I^\varphi + \varphi \left(\frac{1}{\varphi} + b_j\right) \beta_j S^{\varrho_j} I^{\varphi+(\zeta_j-1)} + \frac{1}{2} \varphi(\varphi - 1) \left(\frac{1}{\varphi} + b_j\right) \\ & \times \sigma_j^2 S^{2\varrho_j} I^{\varphi+2(\zeta_j-1)} - 2\vartheta_2(\mu_j + \eta_j) R^2 + 2\vartheta_2\delta_j IR + I^\varphi \sum_{k \in \mathcal{E}, k \neq j} \gamma_{jk}(b_k - b_j). \end{aligned} \quad (11)$$

1. Consider the case where $\zeta_j < 1$. Since $S, I \in (0, 1)$, and $I \leq 1 - S$, then if $\varphi < \min_{j \in \mathcal{E}} \{\zeta_j\}$, (11) yields to

$$\begin{aligned} \mathcal{L}V \leq & -2\vartheta_1\mu_j(1 - S)^2 - 2\vartheta_2(\mu_j + \eta_j) R^2 + 2(\vartheta_2\delta_j - \vartheta_1\eta_j) RI + 2\vartheta_1\beta_j I^\varphi \\ & + \vartheta_1\sigma_j^2 I^\varphi - (1 + b_j\varphi)(\mu_j + \delta_j) I^\varphi + (1 + b_j\varphi)\beta_j S^{\varrho_j} I^{\varphi+(\zeta_j-1)} \\ & - \frac{1}{2}(1 - \varphi)(1 + b_j\varphi)\sigma_j^2 S^{2\varrho_j} I^{\varphi+2(\zeta_j-1)} + I^\varphi \sum_{k \in \mathcal{E}, k \neq j} \gamma_{jk}(b_k - b_j). \end{aligned} \quad (12)$$

For $\varphi < \min \left\{ \min_{j \in \mathcal{E}} \{\zeta_j\}, \frac{1}{\max_{j \in \mathcal{E}} \{|b_j|\}} \right\}$, the inequality (12) becomes

$$\begin{aligned} \mathcal{L}V \leq & -2\vartheta_1\mu_j(1 - S)^2 - 2\vartheta_2(\mu_j + \eta_j) R^2 + 2(\vartheta_2\delta_j - \vartheta_1\eta_j) RI + (1 + b_j\varphi) I^\varphi \\ & \times \left[\frac{\vartheta_1(2\beta_j + \sigma_j^2)}{1 + b_j\varphi} - \frac{1}{2}(1 - \zeta_j)\sigma_j^2 \underbrace{(S^{\varrho_j} I^{(\zeta_j-1)})^2 + \beta_j S^{\varrho_j} I^{(\zeta_j-1)}}_{\leq \frac{\beta_j^2}{2(1-\zeta_j)\sigma_j^2} - (\mu_j + \delta_j)} - (\mu_j + \delta_j) \right] \\ & + \frac{1}{1 + b_j\varphi} \sum_{k \in \mathcal{E}, k \neq j} \gamma_{jk}(b_k - b_j). \end{aligned} \quad (13)$$

Since $\zeta_j < 1$, the quadratic function $-(\mu_j + \delta_j) + \beta_j x - \frac{1}{2}(1 - \zeta_j)\sigma_j^2 x^2$ achieves its maximum value at $x = \frac{\beta_j}{(1 - \zeta_j)\sigma_j^2}$. Consequently, (13) implies

$$\begin{aligned} \mathcal{L}V \leq & -2\vartheta_1\mu_j(1 - S)^2 - 2\vartheta_2(\mu_j + \eta_j) R^2 + 2(\vartheta_2\delta_j - \vartheta_1\eta_j) RI + (1 + b_j\varphi) I^\varphi \\ & \times \left[\frac{\vartheta_1(2\beta_j + \sigma_j^2)}{1 + b_j\varphi} + \frac{\beta_j^2}{2(1 - \zeta_j)\sigma_j^2} - (\mu_j + \delta_j) \right] \end{aligned}$$

$$+ \frac{1}{1 + b_j\varphi} \sum_{k \in \mathcal{E}, k \neq j} \gamma_{jk}(b_k - b_j) \Big]. \quad (14)$$

In accordance with the definition of $\tilde{\Lambda}_j$ in (7), form (14), we have

$$\begin{aligned} \mathcal{L}V \leq & -2\vartheta_1\mu_j(1 - S)^2 - 2\vartheta_2(\mu_j + \eta_j) R^2 + 2(\vartheta_2\delta_j - \vartheta_1\eta_j) RI + (1 + b_j\varphi) I^\varphi \\ & \times \left[\frac{\vartheta_1(2\beta_j + \sigma_j^2)}{1 + b_j\varphi} + \tilde{\Lambda}_j - \frac{b_j\varphi}{1 + b_j\varphi} \sum_{k \in \mathcal{E}, k \neq j} \gamma_{jk}(b_k - b_j) \right. \\ & \left. + \sum_{k \in \mathcal{E}, k \neq j} \gamma_{jk}(b_k - b_j) \right]. \end{aligned} \quad (15)$$

Since the generator matrix Γ is irreducible, there exists a vector $B = (b_1, \dots, b_r)^\top$ that satisfies the Poisson system associated with $\tilde{\Lambda} = (\tilde{\Lambda}_1, \dots, \tilde{\Lambda}_r)^\top$ (see [37] for more details).

$$\Gamma B = -\tilde{\Lambda} + \left(\sum_{j=1}^r \pi_j \tilde{\Lambda}_j \right) \mathbf{e}, \quad (16)$$

where $\mathbf{e}$ denotes the column vector with all entries equal to 1. By substituting Eq. (16) into inequality (15), we obtain

$$\begin{aligned} \mathcal{L}V \leq & -2\vartheta_1\mu_j(1 - S)^2 - 2\vartheta_2(\mu_j + \eta_j) R^2 + 2(\vartheta_2\delta_j - \vartheta_1\eta_j) IR + (1 + b_j\varphi) I^\varphi \\ & \times \left[\frac{\vartheta_1(2\beta_j + \sigma_j^2)}{1 + b_j\varphi} + \frac{b_j\varphi}{1 + b_j\varphi} \left(\tilde{\Lambda}_j - \sum_{j=1}^r \pi_j \tilde{\Lambda}_j \right) + \sum_{j=1}^r \pi_j \tilde{\Lambda}_j \right]. \end{aligned}$$

By (9), we can choose a sufficiently small value for φ_0 such that

$$\max_{j \in \mathcal{E}} \left\{ \frac{b_j\varphi_0}{1 + b_j\varphi_0} \left(\tilde{\Lambda}_j - \sum_{j=1}^r \pi_j \tilde{\Lambda}_j \right) + \sum_{j=1}^r \pi_j \tilde{\Lambda}_j \right\} < 0.$$

Next, we choose φ , ϑ_1 , and ϑ_2 satisfying the following conditions

$$\varphi < \min \left\{ \frac{1}{\max_{j \in \mathcal{E}} \{|b_j|\}}, \min_{j \in \mathcal{E}} \{\zeta_j\}, \varphi_0 \right\},$$

$$\begin{aligned} 0 < \vartheta_1 < & - \frac{(1 + \varphi \min_{j \in \mathcal{E}} \{b_j\})}{2 \max_{j \in \mathcal{E}} \{\beta_j\} + \max_{j \in \mathcal{E}} \{\sigma_j^2\}} \times \max_{j \in \mathcal{E}} \left\{ \frac{b_j\varphi}{1 + b_j\varphi} \left(\tilde{\Lambda}_j - \sum_{j=1}^r \pi_j \tilde{\Lambda}_j \right) \right. \\ & \left. + \sum_{j=1}^r \pi_j \tilde{\Lambda}_j \right\}, \end{aligned}$$

and

$$\vartheta_2 < \min_{j \in \mathcal{E}} \left\{ \frac{\vartheta_1\eta_j}{\delta_j} \right\}.$$

Hence,

$$\mathcal{L}V \leq -(A_j(1 - S)^2 + C_j I^\varphi + B_j R^2), \quad (17)$$

where

$$A_j = 2\vartheta_1\mu_j > 0, \quad B_j = 2\vartheta_2(\mu_j + \eta_j) > 0,$$

and

$$C_j = -(1 + b_j\varphi) \left[\frac{\vartheta_1(2\beta_j + \sigma_j^2)}{1 + b_j\varphi} + \frac{b_j\varphi}{1 + b_j\varphi} \left(\tilde{\Lambda}_j - \sum_{j=1}^r \pi_j \tilde{\Lambda}_j \right) + \sum_{j=1}^r \pi_j \tilde{\Lambda}_j \right] > 0.$$

2. Now, let us consider the case where $\zeta_j \geq 1$. For all $\varphi \leq 1$, given that $S, I \in (0, 1)$ and $I \leq 1 - S$, the expression in (11) can be rewritten as follows

$$\mathcal{L}V \leq -2\vartheta_1\mu_j(1 - S)^2 + 2\vartheta_1\beta_j I^\varphi + \vartheta_1\sigma_j^2 I^\varphi - (1 + b_j\varphi)(\mu_j + \delta_j) I^\varphi$$

$$+ (1 + b_j \varphi) \beta_j S^{\rho_j} I^{\varphi + (\zeta_j - 1)} + \frac{1}{2} \varphi \sigma_j^2 - \frac{1}{2} (1 + b_j \varphi) \sigma_j^2 S^{2\rho_j} I^{\varphi + 2(\zeta_j - 1)} \\ - 2\vartheta_2(\mu_j + \eta_j) R^2 + 2(\vartheta_2 \delta_j - \vartheta_1 \eta_j) I R + I^\varphi \sum_{k \in \mathcal{E}, k \neq j} \gamma_{jk} (b_k - b_j).$$

Rearranging, we obtain

$$\mathcal{L}V \leq -2\vartheta_1 \mu_j (1 - S)^2 - 2\vartheta_2(\mu_j + \eta_j) R^2 + 2(\vartheta_2 \delta_j - \vartheta_1 \eta_j) I R + (1 + b_j \varphi) I^\varphi \\ \times \left(\frac{\vartheta_1 (2\beta_j + \sigma_j^2)}{1 + b_j \varphi} + \frac{1}{2} \varphi \sigma_j^2 + \Theta_j(S^{\rho_j} I^{(\zeta_j - 1)}) \right) \\ \times \left(\frac{1}{1 + b_j \varphi} \sum_{k \in \mathcal{E}, k \neq j} \gamma_{jk} (b_k - b_j) \right). \quad (18)$$

By Lemma 1 and the fact that the quadratic function Θ_j is increasing on the interval $\left[0, \frac{\beta_j}{\sigma_j^2}\right]$ and reaches its maximum at $\frac{\beta_j}{\sigma_j^2}$, the inequality (18) leads to the following two cases

- If $\frac{\beta_j}{\sigma_j^2} \leq \frac{\rho_j^j (\zeta_j - 1)^{\zeta_j - 1}}{(\rho_j + \zeta_j - 1)^{\rho_j + \zeta_j - 1}}$,

$$\mathcal{L}V \leq -2\vartheta_1 \mu_j (1 - S)^2 - 2\vartheta_2(\mu_j + \eta_j) R^2 + 2(\vartheta_2 \delta_j - \vartheta_1 \eta_j) I R \\ + (1 + b_j \varphi) I^\varphi \left(\frac{\vartheta_1 (2\beta_j + \sigma_j^2)}{1 + b_j \varphi} + \frac{1}{2} \varphi \sigma_j^2 + \Theta_j \left(\frac{\beta_j}{\sigma_j^2} \right) \right. \\ \left. + \frac{\sum_{k \in \mathcal{E}, k \neq j} \gamma_{jk} (b_k - b_j)}{1 + b_j \varphi} \right). \quad (19)$$

- If $\frac{\beta_j}{\sigma_j^2} > \frac{\rho_j^j (\zeta_j - 1)^{\zeta_j - 1}}{(\rho_j + \zeta_j - 1)^{\rho_j + \zeta_j - 1}}$,

$$\mathcal{L}V \leq -2\vartheta_1 \mu_j (1 - S)^2 - 2\vartheta_2(\mu_j + \eta_j) R^2 + 2(\vartheta_2 \delta_j - \vartheta_1 \eta_j) I R + (1 + b_j \varphi) I^\varphi \\ \times \left(\frac{\vartheta_1 (2\beta_j + \sigma_j^2)}{1 + b_j \varphi} + \frac{1}{2} \varphi \sigma_j^2 + \Theta_j \left(\frac{\rho_j^j (\zeta_j - 1)^{\zeta_j - 1}}{(\rho_j + \zeta_j - 1)^{\rho_j + \zeta_j - 1}} \right) \right. \\ \left. + \frac{\sum_{k \in \mathcal{E}, k \neq j} \gamma_{jk} (b_k - b_j)}{1 + b_j \varphi} \right). \quad (20)$$

By (19), (20) and the definition of $\bar{\Lambda}_j$, as given in (8), one easily has

$$\mathcal{L}V \leq -2\vartheta_1 \mu_j (1 - S)^2 - 2\vartheta_2(\mu_j + \eta_j) R^2 + 2(\vartheta_2 \delta_j - \vartheta_1 \eta_j) I R + (1 + b_j \varphi) I^\varphi \\ \times \left(\frac{\vartheta_1 (2\beta_j + \sigma_j^2)}{1 + b_j \varphi} + \frac{1}{2} \varphi \sigma_j^2 + \bar{\Lambda}_j - \frac{b_j \varphi}{1 + b_j \varphi} \sum_{k \in \mathcal{E}, k \neq j} \gamma_{jk} (b_k - b_j) \right) \\ + \sum_{k \in \mathcal{E}, k \neq j} \gamma_{jk} (b_k - b_j).$$

Using similar reasoning to the first case (i), and by slightly adjusting the choice of φ and ϑ_1 , it follows that there exist positive real constants A_j , D_j , and B_j . By combining Eqs. (10) and (16), we can deduce

$$\mathcal{L}V \leq -2\vartheta_1 \mu_j (1 - S)^2 - 2\vartheta_2(\mu_j + \eta_j) R^2 + 2(\vartheta_2 \delta_j - \vartheta_1 \eta_j) R I + (1 + b_j \varphi) I^\varphi \\ \times \left[\frac{\vartheta_1 (2\beta_j + \sigma_j^2)}{1 + b_j \varphi} + \frac{1}{2} \varphi \sigma_j^2 + \frac{b_j \varphi}{1 + b_j \varphi} \left(\bar{\Lambda}_j - \sum_{j=1}^r \pi_j \bar{\Lambda}_j \right) + \sum_{j=1}^r \pi_j \bar{\Lambda}_j \right] \\ \triangleq -(A_j (1 - S)^2 + D_j I^\varphi + B_j R^2). \quad (21)$$

Applying Lemma 2 and using inequalities (17) and (21), we conclude the proof of the theorem. $\square$

3.2. Almost sure exponential extinction

Theorem 3. Let $(S_0, I_0, R_0, \xi_0) \in \Delta \times \mathcal{E}$. Then, the disease-free equilibrium state E_0 of system (3) is almost surely exponentially extinct if one of the following conditions holds

1. If $\zeta_j \geq 1$, $\frac{\beta_j}{2\sigma_j^2} > \frac{\rho_j^j (\zeta_j - 1)^{\zeta_j - 1}}{2(\rho_j + \zeta_j - 1)^{\rho_j + \zeta_j - 1}}$, $j \in \mathcal{E}$, and

$$\sum_{j=1}^r \pi_j \tilde{\Psi}_j \left(\frac{\rho_j^j (\zeta_j - 1)^{\zeta_j - 1}}{2(\rho_j + \zeta_j - 1)^{\rho_j + \zeta_j - 1}} \right) < 0, \quad (22)$$

where

$$\tilde{\Psi}_j(x) = -\mu_j + 2\beta_j x - 2\sigma_j^2 x^2, \quad j \in \mathcal{E}, \quad x \in \mathbb{R}.$$

2. If $\zeta_j \geq 1$, $\frac{\beta_j}{2\sigma_j^2} \leq \frac{\rho_j^j (\zeta_j - 1)^{\zeta_j - 1}}{2(\rho_j + \zeta_j - 1)^{\rho_j + \zeta_j - 1}}$, $j \in \mathcal{E}$, and

$$\sum_{j=1}^r \pi_j \left(\frac{\beta_j^2 - 2\mu_j \sigma_j^2}{2\sigma_j^2} \right) < 0. \quad (23)$$

Proof of Theorem 3. For the case where $\zeta_j \geq 1$, define the function $U(S, I, R, j) = \ln(1 - S + I + R)$. By applying Itô's lemma, we obtain

$$dU(S, I, R, j) = \frac{1}{1 - S + I + R} (-\mu_\xi (1 - S) + 2\beta_\xi S^{\rho_\xi} I^{\zeta_\xi} - \mu_\xi I - (\mu_\xi + 2\eta_\xi) R) dt \\ - 2\sigma_\xi^2 \left(\frac{S^{\rho_\xi} I^{\zeta_\xi}}{1 - S + I + R} \right)^2 dt + \frac{2\sigma_\xi S^{\rho_\xi} I^{\zeta_\xi}}{1 - S + I + R} dW. \quad (24)$$

Given that

$$\frac{1}{1 - S + I + R} (-\mu_\xi (1 - S) - \mu_\xi I - (\mu_\xi + 2\eta_\xi) R) \leq -\mu_\xi. \quad (25)$$

From (24) and (25), we deduce that

$$dU(S, I, R, j) \leq \left[-\mu_\xi + 2\beta_\xi \left(\frac{S^{\rho_\xi} I^{\zeta_\xi}}{1 - S + I + R} \right) - 2\sigma_\xi^2 \left(\frac{S^{\rho_\xi} I^{\zeta_\xi}}{1 - S + I + R} \right)^2 \right] dt \\ + \frac{2\sigma_\xi S^{\rho_\xi} I^{\zeta_\xi}}{1 - S + I + R} dW \\ = \tilde{\Psi}_\xi \left(\frac{S^{\rho_\xi} I^{\zeta_\xi}}{1 - S + I + R} \right) dt + \frac{2\sigma_\xi S^{\rho_\xi} I^{\zeta_\xi}}{1 - S + I + R} dW. \quad (26)$$

By integrating (26), we obtain

$$U(t) \leq U(0) + \int_0^t \tilde{\Psi}_\xi(s) \left(\frac{S^{\rho_\xi(s)} I^{\zeta_\xi(s)}}{1 - S(s) + I(s) + R(s)} \right) ds + M_t, \quad (27)$$

where

$$M_t = \int_0^t \frac{2\sigma_\xi(s) (S(s))^{\rho_\xi(s)} (I(s))^{\zeta_\xi(s)}}{1 - S(s) + I(s) + R(s)} dW(s). \quad (28)$$

Since $S, I, R \in (0, 1)$, $I \leq 1 - S$ and $1 - S + I + R = 2(1 - S)$, then by Lemma 1, it follows that

$$\frac{S^{\rho_\xi} I^{\zeta_\xi}}{1 - S + I + R} \leq \frac{1}{2} S^{\rho_\xi} (1 - S)^{\zeta_\xi - 1} \\ \leq \frac{\rho_\xi^{\rho_\xi} (\zeta_\xi - 1)^{(\zeta_\xi - 1)}}{2(\rho_\xi + \zeta_\xi - 1)^{\rho_\xi + \zeta_\xi - 1}}. \quad (29)$$

According to (28) and (29), we obtain

$$[M]_t = \int_0^t \left(\frac{2\sigma_\xi(s) (S(s))^{\rho_\xi(s)} (I(s))^{\zeta_\xi(s)}}{1 - S(s) + I(s) + R(s)} \right)^2 ds \\ \leq \left[\max_{j \in \mathcal{E}} \sigma_j^2 \left(\frac{\rho_j^j (\zeta_j - 1)^{\zeta_j - 1}}{(\rho_j + \zeta_j - 1)^{\rho_j + \zeta_j - 1}} \right)^2 \right] t. \quad (30)$$

Thus, applying the law of large numbers for martingales (see Mao [39]), we obtain

$$\limsup_{t \rightarrow \infty} \frac{M_t}{t} = 0 \quad \text{a.s.} \quad (31)$$

1. If $\frac{\beta_j}{2\sigma_j^2} > \frac{\rho_j^{\rho_j} (\zeta_j - 1)^{\zeta_j - 1}}{2(\rho_j + \zeta_j - 1)^{\rho_j + \zeta_j - 1}}$, then the function $\tilde{\Psi}_j$ increases in $\left[0, \frac{\beta_j}{2\sigma_j^2}\right]$.
From (29), it follows that

$$\tilde{\Psi}_\xi \left(\frac{S^{\rho_\xi} I^{\zeta_\xi}}{1 - S + I + R} \right) \leq \tilde{\Psi}_\xi \left(\frac{\rho_\xi^{\rho_\xi} (\zeta_\xi - 1)^{(\zeta_\xi - 1)}}{2(\rho_\xi + \zeta_\xi - 1)^{\rho_\xi + \zeta_\xi - 1}} \right). \quad (32)$$

Therefore, from (27) and (32), we get

$$U(t) \leq U(0) + \int_0^t \tilde{\Psi}_{\xi(s)} \left(\frac{\rho_{\xi(s)}^{\rho_{\xi(s)}} (\zeta_{\xi(s)} - 1)^{(\zeta_{\xi(s)} - 1)}}{2(\rho_{\xi(s)} + \zeta_{\xi(s)} - 1)^{\rho_{\xi(s)} + \zeta_{\xi(s)} - 1}} \right) ds + M_t. \quad (33)$$

Applying Birkhoff's Ergodic Theorem yields

$$\begin{aligned} \limsup_{t \rightarrow \infty} \frac{1}{t} \int_0^t \tilde{\Psi}_{\xi(s)} \left(\frac{\rho_{\xi(s)}^{\rho_{\xi(s)}} (\zeta_{\xi(s)} - 1)^{(\zeta_{\xi(s)} - 1)}}{2(\rho_{\xi(s)} + \zeta_{\xi(s)} - 1)^{\rho_{\xi(s)} + \zeta_{\xi(s)} - 1}} \right) ds \\ \stackrel{\text{a.s.}}{=} \sum_{j=1}^r \pi_j \tilde{\Psi}_j \left(\frac{\rho_j^{\rho_j} (\zeta_j - 1)^{\zeta_j - 1}}{2(\rho_j + \zeta_j - 1)^{\rho_j + \zeta_j - 1}} \right). \end{aligned} \quad (34)$$

If (22) holds, then, by (31), (33), and (34), we obtain

$$\begin{aligned} \limsup_{t \rightarrow \infty} \frac{1}{t} \ln [1 - S(t) + I(t) + R(t)] &\leq \sum_{j=1}^r \pi_j \tilde{\Psi}_j \left(\frac{\rho_j^{\rho_j} (\zeta_j - 1)^{\zeta_j - 1}}{2(\rho_j + \zeta_j - 1)^{\rho_j + \zeta_j - 1}} \right) \\ &< 0 \quad \text{a.s.} \end{aligned} \quad (35)$$

2. If $\frac{\beta_j}{2\sigma_j^2} \leq \frac{\rho_j^{\rho_j} (\zeta_j - 1)^{\zeta_j - 1}}{2(\rho_j + \zeta_j - 1)^{\rho_j + \zeta_j - 1}}$, then from (27) and the fact that the function $\tilde{\Psi}_j$ attains its maximum at $\frac{\beta_j}{2\sigma_j^2}$, one can easily verify

$$U(t) \leq U(0) + \int_0^t \tilde{\Psi}_{\xi(s)} \left(\frac{\beta_{\xi(s)}}{2\sigma_{\xi(s)}^2} \right) ds + \int_0^t \frac{2\sigma_{\xi(s)}^{\rho_{\xi(s)}} S^{\rho_{\xi(s)}} I^{\zeta_{\xi(s)}}}{1 - S(s) + I(s) + R(s)} dW(s), \quad (36)$$

where

$$\tilde{\Psi}_{\xi(s)} \left(\frac{\beta_{\xi(s)}}{2\sigma_{\xi(s)}^2} \right) = \frac{\beta_{\xi(s)}^2 - 2\mu_{\xi(s)} \sigma_{\xi(s)}^2}{2\sigma_{\xi(s)}^2}.$$

Similarly to the proof (i), if (23) holds, then

$$\begin{aligned} \limsup_{t \rightarrow \infty} \frac{1}{t} \ln [1 - S(t) + I(t) + R(t)] &\leq \sum_{j=1}^r \pi_j \left(\frac{\beta_j^2 - 2\mu_j \sigma_j^2}{2\sigma_j^2} \right) \\ &< 0 \quad \text{a.s.} \end{aligned} \quad (37)$$

Using (35), and (37), we derive

$$\limsup_{t \rightarrow \infty} \frac{1}{t} \ln [1 - S(t) + I(t) + R(t)] < 0, \quad \text{a.s.}$$

This completes the proof. $\square$

3.3. Exponential moment stability

We present the following theorem, which establishes sufficient conditions to ensure the exponential stability of the q -th moment of the disease-free equilibrium state E_0 for the system (3).

Theorem 4. Let $q > 0$. The solution $(S(t), I(t), R(t), \xi(t))$ of the stochastic differential system (3) is said to be exponentially stable in the q -th moment if one of the following conditions holds

1. If $\zeta_j \geq 1$, $\frac{\beta_j}{2\sigma_j^2} > \frac{\rho_j^{\rho_j} (\zeta_j - 1)^{\zeta_j - 1}}{(\rho_j + \zeta_j - 1)^{\rho_j + \zeta_j - 1}}$, for all $j \in \mathcal{E}$, and

$$\sum_{j=1}^r \pi_j \left[-\mu_j + 2\beta_j - \frac{\rho_j^{\rho_j} (\zeta_j - 1)^{\zeta_j - 1}}{(\rho_j + \zeta_j - 1)^{\rho_j + \zeta_j - 1}} + 2(q-1)\sigma_j^2 \left(\frac{\rho_j^{\rho_j} (\zeta_j - 1)^{\zeta_j - 1}}{(\rho_j + \zeta_j - 1)^{\rho_j + \zeta_j - 1}} \right)^2 \right] < 0. \quad (38)$$

2. If $\zeta_j \geq 1$, $\frac{\beta_j}{2\sigma_j^2} \leq \frac{\rho_j^{\rho_j} (\zeta_j - 1)^{\zeta_j - 1}}{(\rho_j + \zeta_j - 1)^{\rho_j + \zeta_j - 1}}$, for all $j \in \mathcal{E}$, and

$$\sum_{j=1}^r \pi_j \left[\frac{\beta_j^2 - 2\mu_j \sigma_j^2}{2\sigma_j^2} + \frac{1}{2} q \sigma_j^2 \left(\frac{\rho_j^{\rho_j} (\zeta_j - 1)^{\zeta_j - 1}}{(\rho_j + \zeta_j - 1)^{\rho_j + \zeta_j - 1}} \right)^2 \right] < 0. \quad (39)$$

Proof of Theorem 4.

1. For the case where $\zeta_j \geq 1$ and $\frac{\beta_j}{2\sigma_j^2} > \frac{\rho_j^{\rho_j} (\zeta_j - 1)^{\zeta_j - 1}}{2(\rho_j + \zeta_j - 1)^{\rho_j + \zeta_j - 1}}$, one can easily derive from (33)

$$\begin{aligned} \ln [1 - S(t) + I(t) + R(t)]^q &\leq \ln [(1 - S_0 + I_0 + R_0)^q] + qM_t \\ &\quad + q \int_0^t \tilde{\Psi}_{\xi(s)} \left(\frac{\rho_{\xi(s)}^{\rho_{\xi(s)}} (\zeta_{\xi(s)} - 1)^{(\zeta_{\xi(s)} - 1)}}{2(\rho_{\xi(s)} + \zeta_{\xi(s)} - 1)^{\rho_{\xi(s)} + \zeta_{\xi(s)} - 1}} \right) ds. \end{aligned} \quad (40)$$

By Birkhoff's Ergodic Theorem, for any $\varepsilon > 0$ and for sufficiently large values of t , we have

$$\begin{aligned} \int_0^t \tilde{\Psi}_{\xi(s)} \left(\frac{\rho_{\xi(s)}^{\rho_{\xi(s)}} (\zeta_{\xi(s)} - 1)^{(\zeta_{\xi(s)} - 1)}}{2(\rho_{\xi(s)} + \zeta_{\xi(s)} - 1)^{\rho_{\xi(s)} + \zeta_{\xi(s)} - 1}} \right) ds \\ \leq \left[\sum_{j=1}^r \pi_j \tilde{\Psi}_j \left(\frac{\rho_j^{\rho_j} (\zeta_j - 1)^{(\zeta_j - 1)}}{2(\rho_j + \zeta_j - 1)^{\rho_j + \zeta_j - 1}} \right) + \frac{\varepsilon}{2q} \right] t. \end{aligned} \quad (41)$$

By combining (40) and (41), we obtain

$$\mathbb{E} \left[\frac{(1 - S(t) + I(t) + R(t))^q}{(1 - S_0 + I_0 + R_0)^q} \right] \leq e \left[\sum_{j=1}^r \pi_j \tilde{\Psi}_j \left(\frac{\rho_j^{\rho_j} (\zeta_j - 1)^{(\zeta_j - 1)}}{2(\rho_j + \zeta_j - 1)^{\rho_j + \zeta_j - 1}} \right) + \frac{\varepsilon}{2q} \right] \mathbb{E} [e^{qM_t}]. \quad (42)$$

On the one hand, qM_t is a continuous real-valued martingale with the initial condition $qM_0 = 0$. Moreover, according to (29) and (30), the quadratic variation obeys

$$[qM]_t \leq 4q^2 \int_0^t \sigma_{\xi(s)}^2 \left(\frac{\rho_{\xi(s)}^{\rho_{\xi(s)}} (\zeta_{\xi(s)} - 1)^{(\zeta_{\xi(s)} - 1)}}{2(\rho_{\xi(s)} + \zeta_{\xi(s)} - 1)^{\rho_{\xi(s)} + \zeta_{\xi(s)} - 1}} \right)^2 ds. \quad (43)$$

By the ergodic property of the Markov chain $\xi(t)$ and (43), it follows that for sufficiently large t , we have

$$[qM]_t \leq q^2 \left[\sum_{j=1}^r 4\pi_j \sigma_j^2 \left(\frac{\rho_j^{\rho_j} (\zeta_j - 1)^{(\zeta_j - 1)}}{2(\rho_j + \zeta_j - 1)^{\rho_j + \zeta_j - 1}} \right)^2 + \frac{\varepsilon}{q^2} \right] t.$$

On the other hand, $\exp \left(qM_t - \frac{1}{2} [qM]_t \right)$ is a martingale. Consequently, for sufficiently large values of t , we deduce that

$$\begin{aligned} \mathbb{E} [e^{qM_t}] &= \mathbb{E} \left[e^{\frac{1}{2} [qM]_t} \right] \\ &\leq e \left[\sum_{j=1}^r 4\pi_j \sigma_j^2 \left(\frac{\rho_j^{\rho_j} (\zeta_j - 1)^{(\zeta_j - 1)}}{2(\rho_j + \zeta_j - 1)^{\rho_j + \zeta_j - 1}} \right)^2 + \frac{\varepsilon}{2q^2} \right] t. \end{aligned} \quad (44)$$

By combining (42) and (44), we get

$$\begin{aligned} \mathbb{E} \left[\frac{(1 - S(t) + I(t) + R(t))^q}{(1 - S_0 + I_0 + R_0)^q} \right] \\ \leq e \left[\sum_{j=1}^r \pi_j \left[\tilde{\Psi}_j \left(\frac{\rho_j^{\rho_j} (\zeta_j - 1)^{(\zeta_j - 1)}}{2(\rho_j + \zeta_j - 1)^{\rho_j + \zeta_j - 1}} \right) + 2q \sigma_j^2 \left(\frac{\rho_j^{\rho_j} (\zeta_j - 1)^{(\zeta_j - 1)}}{2(\rho_j + \zeta_j - 1)^{\rho_j + \zeta_j - 1}} \right)^2 \right] t + \varepsilon t \right] \\ = e \left[\sum_{j=1}^r \pi_j \left[-\mu_j + 2\beta_j - \frac{\rho_j^{\rho_j} (\zeta_j - 1)^{(\zeta_j - 1)}}{2(\rho_j + \zeta_j - 1)^{\rho_j + \zeta_j - 1}} + 2(q-1)\sigma_j^2 \left(\frac{\rho_j^{\rho_j} (\zeta_j - 1)^{(\zeta_j - 1)}}{2(\rho_j + \zeta_j - 1)^{\rho_j + \zeta_j - 1}} \right)^2 \right] t + \varepsilon t \right]. \end{aligned} \quad (45)$$

Thus, if (38) holds, then by letting $t \rightarrow +\infty$ and $\varepsilon \rightarrow 0$, the desired assertion is established.

2. Now, for the case where $\zeta_j \geq 1$ and $\frac{\beta_j}{2\sigma_j^2} \leq \frac{\rho_j^{\rho_j}(\zeta_j-1)^{\zeta_j-1}}{2(\rho_j+\zeta_j-1)^{\rho_j+\zeta_j-1}}$, it follows from (36)

$$\ln[(1-S(t)+I(t)+R(t))^q] \leq \ln[(1-S_0+I_0+R_0)^q] + qM_t + q \int_0^t \left(\frac{\beta_{\xi(s)}^2 - 2\mu_{\xi(s)}\sigma_{\xi(s)}^2}{2\sigma_{\xi(s)}^2} \right) ds.$$

Similarly to the proof (i) and using (39), we get

$$\limsup_{t \rightarrow \infty} \frac{1}{t} \ln \mathbb{E}[(1-S(t)+I(t)+R(t))^q] \leq q \sum_{j=1}^r \pi_j \left(\frac{\beta_j^2 - 2\mu_j\sigma_j^2}{2\sigma_j^2} + \frac{q}{2} \sigma_j^2 \left(\frac{\rho_j^{\rho_j}(\zeta_j-1)^{\zeta_j-1}}{(\rho_j+\zeta_j-1)^{\rho_j+\zeta_j-1}} \right)^2 \right) < 0. \quad (46)$$

This completes the proof. $\square$

4. Stochastic persistence

Stochastic persistence criteria are derived in terms of model parameters 12, 13, 14. The proof relies on the following lemma.

Lemma 3. Let X be a random variable and let $\alpha_0 > 0$ and $C_1 > 0$ be positive constants. Assume that

$$\mathbb{E}[\exp(\alpha_0 X)] + \mathbb{E}[\exp(-\alpha_0 X)] \leq C_1.$$

Then, the logarithmic Laplace transform $\varphi(\alpha) = \ln \mathbb{E}[\exp(\alpha X)]$ is twice differentiable on $[0, \frac{\alpha_0}{2})$, and there exists $C_2 > 0$ such that

$$\varphi(\alpha) \leq \alpha \mathbb{E}[X] + \alpha^2 C_2, \quad \text{for } \alpha \in [0, \frac{\alpha_0}{2}).$$

Proof. The proof of this lemma can be found in [40]. $\square$

Using Lemma 3 and the methodologies described in [40,41], we proceed to demonstrate the stochastic persistence of the disease when $\Lambda > 0$.

Theorem 5. If $\Lambda > 0$, the disease persists stochastically, meaning that for any $\varepsilon > 0$, there exists a constant $v > 0$ such that

$$\liminf_{t \rightarrow \infty} \mathbb{P}_{X_0, j}(I(t) \geq v) > 1 - \varepsilon, \quad \text{for all } (X_0, j) \in \Delta \times \mathcal{E}.$$

Proof of Theorem 5. Consider the function $V_\alpha(S, I, R, j) = I^\alpha$, where $\alpha \in (0, 1)$ is a real constant. The infinitesimal generator $\mathcal{L}V_\alpha$ is given by

$$\begin{aligned} \mathcal{L}V_\alpha &= \alpha I^{\alpha-1} [-(\mu_j + \delta_j)I + \beta_j S^{\rho_j} I^{\zeta_j}] + \frac{1}{2} \alpha(\alpha-1) I^{\alpha-2} (\sigma_j S^{\rho_j} I^{\zeta_j})^2 \\ &= \alpha [-(\mu_j + \delta_j) + \beta_j S^{\rho_j} I^{\zeta_j-1} + \frac{\alpha-1}{2} \sigma_j^2 S^{2\rho_j} I^{2(\zeta_j-1)}] I^\alpha. \end{aligned} \quad (47)$$

We define

$$M_\alpha = \sup_{(S, I, R, j) \in \Delta \times \mathcal{E}} \left\{ \alpha \left(-(\mu_j + \delta_j) + \beta_j S^{\rho_j} I^{\zeta_j-1} + \frac{\alpha-1}{2} \sigma_j^2 S^{2\rho_j} I^{2(\zeta_j-1)} \right) \right\}. \quad (48)$$

Combining (47) and (48), we obtain

$$\mathcal{L}V_\alpha \leq M_\alpha V_\alpha. \quad (49)$$

For any $(X_0, j) = (S_0, I_0, R_0, j) \in \Delta \times \mathcal{E}$, applying Dynkin's formula in conjunction with (49), we get

$$\mathbb{E}^{X_0, j}[I^\alpha(t)] \leq I_0^\alpha + M_\alpha \int_0^t \mathbb{E}^{X_0, j}[I^\alpha(s)] ds.$$

Using Gronwall's inequality, we can establish that

$$\mathbb{E}^{X_0, j}[I^\alpha(t)] \leq I_0^\alpha e^{M_\alpha t}, \quad \text{for } (S_0, I_0, R_0, j) \in \Delta \times \mathcal{E}. \quad (50)$$

Analogously, for any $t \geq pL$ where $p \in \mathbb{N}$ and $L > 0$, we have

$$\mathbb{E}^{X_0, j}[I^\alpha(t)] \leq \mathbb{E}^{X_0, j}[I^\alpha(pL)] e^{M_\alpha(t-pL)}. \quad (51)$$

On the other hand, we have

$$\begin{aligned} \ln(I_0) - \ln(I(t)) &= - \int_0^t \Theta_{\xi(s)} \left((S(s))^{\rho_{\xi(s)}} (I(s))^{\zeta_{\xi(s)}-1} \right) ds \\ &\quad - \int_0^t \sigma_{\xi(s)} (S(s))^{\rho_{\xi(s)}} (I(s))^{\zeta_{\xi(s)}-1} dW(s). \end{aligned} \quad (52)$$

From the ergodicity of the Markov chain $\xi(s)$, we derive

$$\lim_{t \rightarrow \infty} \frac{1}{t} \int_0^t \Theta_{\xi(s)} \left((S(s))^{\rho_{\xi(s)}} (I(s))^{\zeta_{\xi(s)}-1} \right) ds = \Lambda. \quad (53)$$

Applying the dominated convergence theorem and (53), one can easily verify

$$\lim_{t \rightarrow \infty} \mathbb{E} \left[\frac{1}{t} \int_0^t \Theta_{\xi(s)} \left((S(s))^{\rho_{\xi(s)}} (I(s))^{\zeta_{\xi(s)}-1} \right) ds \right] = \Lambda,$$

this implies that there exists $T > 0$ such that

$$\mathbb{E} \left[\frac{1}{T} \int_0^T \Theta_{\xi(s)} \left((S(s))^{\rho_{\xi(s)}} (I(s))^{\zeta_{\xi(s)}-1} \right) ds \right] \geq \frac{\Lambda}{2}. \quad (54)$$

By (52) and (54), we have

$$\begin{aligned} \mathbb{E}^{X_0, j} \left[\ln \left(\frac{I_0}{I(T)} \right) \right] &= -\mathbb{E}^{X_0, j} \left[\int_0^T \Theta_{\xi(s)} \left((S(s))^{\rho_{\xi(s)}} (I(s))^{\zeta_{\xi(s)}-1} \right) ds \right] \\ &\leq -\frac{\Lambda}{2} T. \end{aligned} \quad (55)$$

From (50), we obtain

$$\begin{aligned} \mathbb{E}^{X_0, j} \left[e^{\ln \left(\frac{I_0}{I(T)} \right)} + e^{\ln \left(\frac{I(T)}{I_0} \right)} \right] &\leq e^{M_{-1}T} + e^{M_1T} \\ &< \infty, \quad \text{for any } T \geq 0. \end{aligned}$$

Applying Lemma (3) and (55), we deduce

$$\ln \left(\mathbb{E}^{X_0, j} \left[e^{\alpha \ln \left(\frac{I_0}{I(T)} \right)} \right] \right) \leq -\frac{\Lambda \alpha}{2} T + \widehat{M} \alpha^2, \quad \text{for } \alpha \in [0, 0.5]. \quad (56)$$

where $\widehat{M}$ is a constant depending on T , M_{-1} , and M_1 . For sufficiently small α , inequality (56), for $I_0 > \varrho$, $j \in \mathcal{E}$ yields

$$\begin{aligned} \mathbb{E}^{X_0, j} \left[\frac{I_0^\alpha}{I^\alpha(T)} \right] &= \mathbb{E}^{X_0, j} \left[e^{\alpha \ln \left(\frac{I_0}{I(T)} \right)} \right] \\ &\leq e^{\left(-\frac{\Lambda \alpha}{2} T + \widehat{M} \alpha^2 \right)} \\ &\leq e^{\left(-\frac{\Lambda \alpha}{4} T \right)}. \end{aligned}$$

For a sufficiently small $\varrho > 0$ such that $I_0 > \varrho$, we have

$$\mathbb{E}^{X_0, j} [I^{-\alpha}(T)] \leq \kappa I_0^{-\alpha}, \quad \text{where } \kappa = \exp \left(-\frac{\Lambda \alpha}{4} T \right). \quad (57)$$

Inequalities (50) and (57) imply that

$$\begin{aligned} \mathbb{E}^{X_0, j} [I^{-\alpha}(T)] &\leq \kappa I_0^{-\alpha} + C \\ &= \kappa \mathbb{E}^{X_0, j} [I_0^{-\alpha}] + C, \end{aligned} \quad (58)$$

where $C = \varrho^{-\alpha} \exp(M_{-a}T)$. Applying the Markov property and (58), we conclude that

$$\mathbb{E}^{X_0, j} [I^{-\alpha}\{(p+1)T\}] \leq \kappa \mathbb{E}^{X_0, j} [I^{-\alpha}(pT)] + C, \quad \text{for } p \in \mathbb{N}.$$

By iterating the previous inequality, we derive

$$\mathbb{E}^{X_0, j} [I^{-\alpha}(nT)] \leq \kappa^n I_0^{-\alpha} + \frac{C(1-\kappa^n)}{1-\kappa}, \quad \text{for } n \in \mathbb{N}. \quad (59)$$

From (51) and (59), we have

$$\mathbb{E}^{X_0, j} [I^{-\alpha}(t)] \leq \left(\kappa^n I_0^{-\alpha} + \frac{C(1-\kappa^n)}{1-\kappa} \right) e^{M_{-a}T}, \quad \text{for } t \in [nT, (n+1)T]. \quad (60)$$

Since $\kappa < 1$, taking $n \rightarrow +\infty$ in (60), yields

$$\begin{aligned} \limsup_{t \rightarrow +\infty} \mathbb{E}^{X_0, j} [I^{-\alpha}(t)] &= \frac{C}{1-\kappa} e^{M_{-a}T} \\ &\triangleq \Upsilon. \end{aligned}$$

Applying Chebyshev's inequality, we obtain

$$\begin{aligned}\mathbb{P}(|I(t)| < v) &= \mathbb{P}\left(\frac{1}{|I(t)|^\alpha} > \frac{1}{v^\alpha}\right) \\ &\leq v^\alpha \mathbb{E}[|I^{-\alpha}(t)|],\end{aligned}$$

for $v = \left(\frac{\epsilon}{Y}\right)^{\frac{1}{\alpha}}$ and an arbitrarily small constant $\epsilon > 0$, we get

$$\limsup_{t \rightarrow \infty} \mathbb{P}(|I(t)| < v) \leq v^\alpha Y = \epsilon.$$

Thus, $\liminf_{t \rightarrow \infty} \mathbb{P}(I(t) \geq v) > 1 - \epsilon$. The proof is complete. $\square$

5. Computer simulations

In this section, we present extensive numerical simulations to validate the theoretical results established in [Theorems 2, 3, and 5](#). Unlike the single-sample path simulations presented in the original manuscript, we now provide ensemble statistics based on $N = 500$ independent simulations for each parameter set. All simulations were implemented in Python 3.10 using the NumPy and Matplotlib libraries. The random seed was fixed to 42 to ensure full reproducibility.

5.1. Numerical scheme and positivity preservation

The standard Euler-Maruyama discretization may produce negative values and violate the invariance of the simplex Δ , compromising the biological relevance of the numerical solution. To address this issue, we employ a balanced positivity-preserving Euler-Maruyama scheme. Let $\Delta t > 0$ be the time step and let k denote the time index. The discretized system is given by

$$\begin{cases} S_{k+1} = \frac{S_k + \alpha_S(S_k, I_k, R_k)\Delta t}{1 + \beta_S(S_k, I_k, R_k)\Delta t} - \frac{\sigma_k S_k^{\rho_k} I_k^{\zeta_k} \Delta W_k}{1 + \beta_S(S_k, I_k, R_k)\Delta t}, \\ I_{k+1} = \frac{I_k + \alpha_I(S_k, I_k, R_k)\Delta t}{1 + \beta_I(S_k, I_k, R_k)\Delta t} + \frac{\sigma_k S_k^{\rho_k} I_k^{\zeta_k} \Delta W_k}{1 + \beta_I(S_k, I_k, R_k)\Delta t}, \\ R_{k+1} = \frac{R_k + \alpha_R(S_k, I_k, R_k)\Delta t}{1 + \beta_R(S_k, I_k, R_k)\Delta t}, \end{cases} \quad (61)$$

where $\Delta W_k \sim \mathcal{N}(0, \Delta t)$ represents the increment of the Wiener process, and the functions α and β are defined as

$$\begin{cases} \alpha_S(S, I, R) = \mu_k - \mu_k S - \beta_S S^{\rho_k} I^{\zeta_k} + \eta_k R, \\ \alpha_I(S, I, R) = -(\mu_k + \delta_k)I + \beta_S S^{\rho_k} I^{\zeta_k}, \\ \alpha_R(S, I, R) = -(\mu_k + \eta_k)R + \delta_k I, \\ \beta_S(S, I, R) = \max\{0, -\alpha_S(S, I, R)/S\}, \\ \beta_I(S, I, R) = \max\{0, -\alpha_I(S, I, R)/I\}, \\ \beta_R(S, I, R) = \max\{0, -\alpha_R(S, I, R)/R\}. \end{cases} \quad (62)$$

The implicit correction terms $\beta_S, \beta_I, \beta_R$ ensure that the denominators remain positive, thereby preventing negative values. After each step, we perform a projection onto the simplex to correct any numerical deviations:

$$(S_{k+1}, I_{k+1}, R_{k+1}) \leftarrow \frac{(S_{k+1}, I_{k+1}, R_{k+1})}{S_{k+1} + I_{k+1} + R_{k+1}}. \quad (63)$$

This scheme guarantees that $S_k, I_k, R_k \in (0, 1)$ and $S_k + I_k + R_k = 1$ for all k whenever the initial condition satisfies these constraints, thus preserving both positivity and the invariance of Δ .

5.2. Convergence analysis

To validate the numerical scheme, we performed a convergence analysis with decreasing time steps $\Delta t \in \{0.1, 0.05, 0.01, 0.005, 0.001\}$. [Fig. 2](#) presents the convergence analysis results.

[Table 1](#) reports the mean squared error (MSE) of the infected population I at $t = 100$ relative to the solution obtained with $\Delta t = 10^{-4}$, averaged over 100 independent simulations for Case g ($\Lambda > 0$).

Based on these results, we selected $\Delta t = 0.05$ and $T_{\text{final}} = 500$ for all subsequent simulations, as it provides an excellent balance between accuracy and computational efficiency.

Table 1

Convergence analysis of the numerical scheme.

Δt	MSE for $I(t)$	Computational time (s)
0.1	2.34×10^{-3}	0.12
0.05	5.67×10^{-4}	0.25
0.01	2.15×10^{-5}	1.18
0.005	5.23×10^{-6}	2.34
0.001	1.02×10^{-7}	11.72

5.3. Reproducibility details

The following information is provided to ensure full reproducibility of the numerical experiments:

- **Random seed:** All simulations use the fixed seed `np.random.seed(42)`.
- **Number of simulations:** $N = 500$ independent trajectories for each parameter set.
- **Time discretization:** $\Delta t = 0.05$, $T_{\text{final}} = 500$.
- **Markov chain generation:** The Markov chain paths are generated using the transition probability matrices given in [Examples 5.1 and 5.2](#).
- **Code availability:** The complete Python code is available as supplementary material.

5.4. Test case validation

To validate the numerical implementation, we first performed a test case with synthetic data. [Fig. 3](#) shows the ensemble statistics for this test case.

5.5. Two-state Markov chains

Consider a continuous-time Markov chain $(\xi(t))_{t \geq 0}$ with the state space $\mathcal{E} = \{e_1, e_2\}$, the generator matrix Γ and the stationary distribution π . For a small time step $\varsigma > 0$, the transition matrix A can be approximated as $A = e^{\varsigma \Gamma}$. Subsequently, we consider the two following Markov chains [4, 5, 6](#).

Example 5.1. Setting $\varsigma = 0.01$ and $\Gamma = \begin{pmatrix} -3 & 3 \\ 4 & -4 \end{pmatrix}$, yields

$$A = \begin{pmatrix} 0.9704 & 0.0296 \\ 0.0408 & 0.9592 \end{pmatrix}, \quad \pi = (\pi_1, \pi_2) = (0.5714, 0.4286).$$

Example 5.2. Setting $\varsigma = 0.0001$ and $\Gamma = \begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix}$, yields

$$A = \begin{pmatrix} 0.9999 & 0.0001 \\ 0.0002 & 0.9998 \end{pmatrix}, \quad \pi = (\pi_1, \pi_2) = (0.6667, 0.3333).$$

5.6. Statistical results for extinction ($\Lambda < 0$)

Case: $\bar{\Lambda}_1 < 0$, $\bar{\Lambda}_2 < 0$, $\bar{\Lambda}_1 < 0$, and $\bar{\Lambda}_2 < 0$

1. Let

$$\begin{aligned}(S_0, I_0, R_0) &= (0.35, 0.55, 0.1), (\zeta_1, \zeta_2) = (0.5, 0.3), (\rho_1, \rho_2) = (0.5, 0.5), \\ (\mu_1, \mu_2) &= (0.999, 0.887), \\ (\beta_1, \beta_2) &= (0.3, 0.4), (\eta_1, \eta_2) = (0.5, 0.5), (\delta_1, \delta_2) = (0.9899, 0.988), \\ (\sigma_1, \sigma_2) &= (0.9999, 0.8899).\end{aligned}$$

Thus,

$$\bar{\Lambda}_1 \approx -1.8989 < 0, \quad \bar{\Lambda}_2 \approx -1.875 < 0,$$

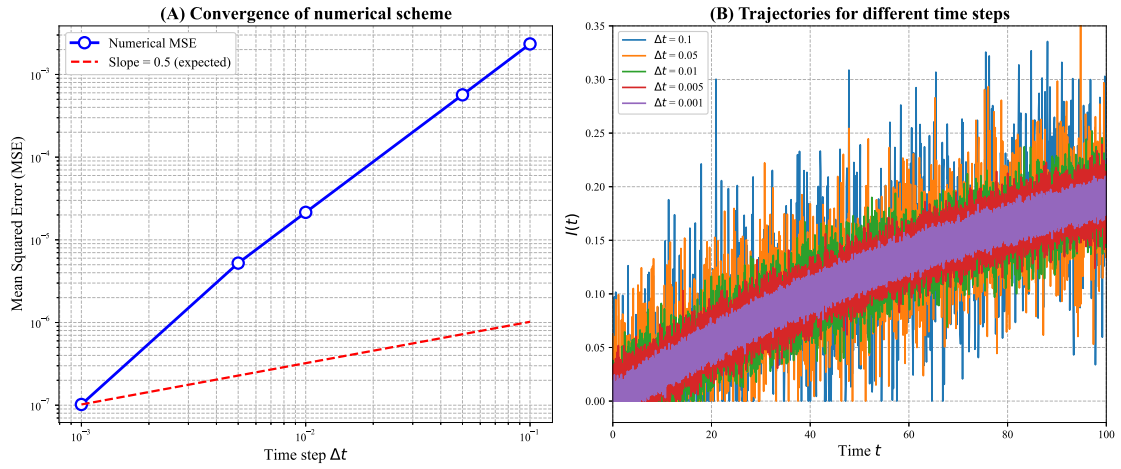


Fig. 2. Convergence analysis of the numerical scheme: (A) Mean squared error (MSE) as a function of the time step Δt on a log-log scale, showing the expected convergence rate of order 0.5. (B) Sample trajectories for different time steps illustrating the numerical stability of the scheme.

Test Case

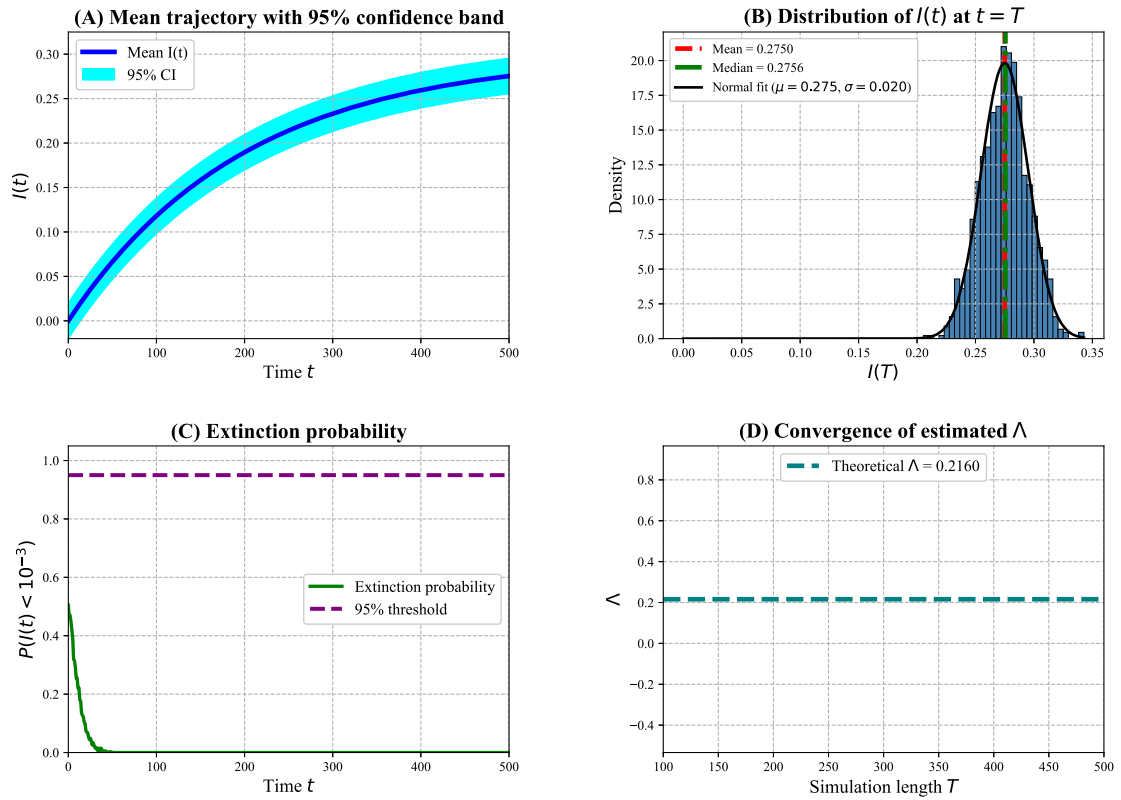


Fig. 3. Test case validation: Ensemble statistics with synthetic data demonstrating the functionality of the numerical implementation.

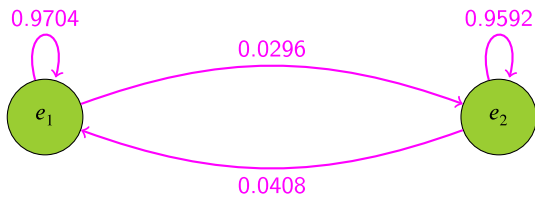


Fig. 4. Two-state Markov chain diagram of Example 5.1.

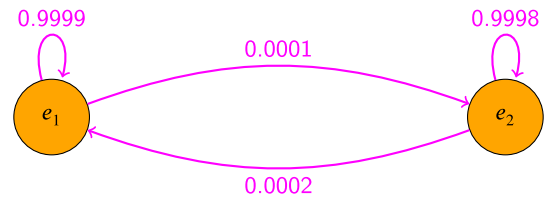


Fig. 5. Two-state Markov chain diagram of Example 5.2.

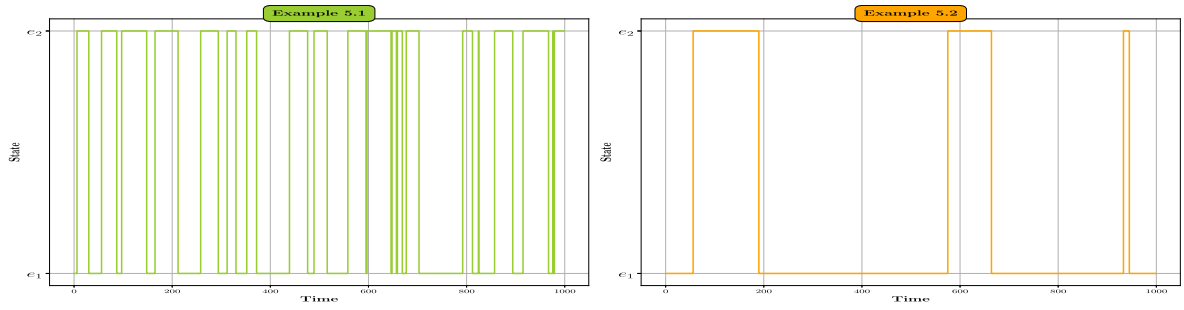
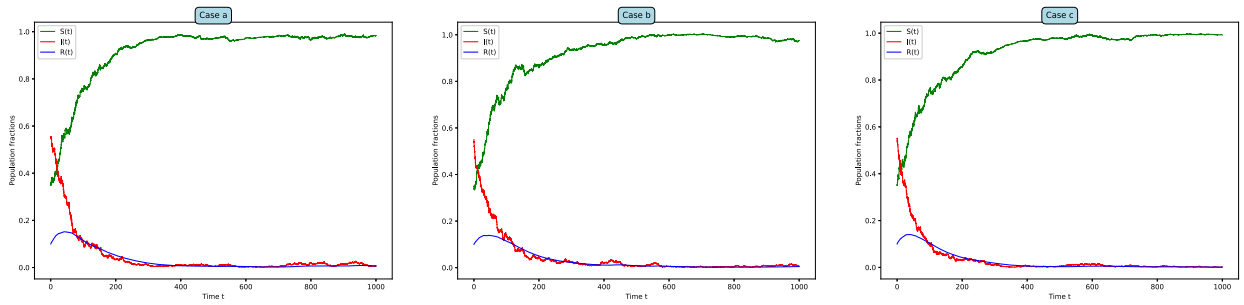
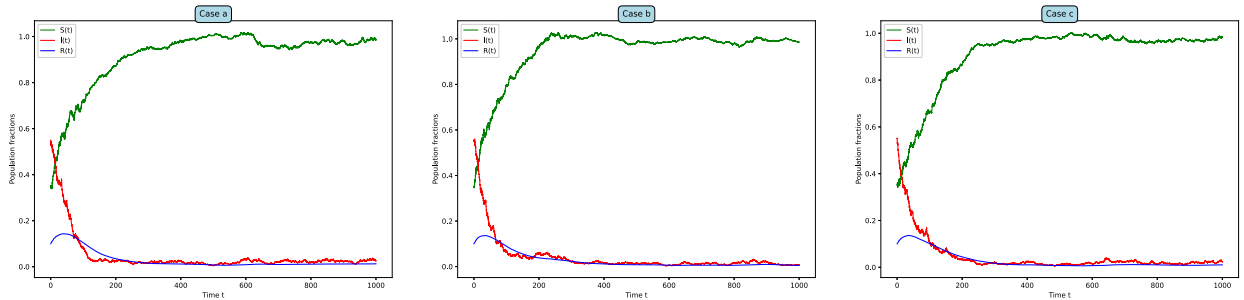


Fig. 6. Transition dynamics of Markov chains.

Table 2
Statistical results for case a (Extinction).

Quantity	Example 5.1	Example 5.2	Theoretical
Λ	-1.85681	-1.84213	-1.85681
$\hat{\Lambda}$ (estimated)	-1.8592 ± 0.0118	-1.8445 ± 0.0123	—
Extinction probability at $T = 500$	0.987	0.985	1
Mean of $I(500)$	3.21×10^{-4}	3.45×10^{-4}	0
95% CI for mean $I(500)$	$[2.89 \times 10^{-4}, 3.53 \times 10^{-4}]$	$[3.12 \times 10^{-4}, 3.78 \times 10^{-4}]$	$[0, 0]$

Fig. 7. Evolution of $S(t)$, $I(t)$, and $R(t)$ in cases a, b, and c, illustrated with Example 5.1.Fig. 8. Evolution of $S(t)$, $I(t)$, and $R(t)$ in cases a, b, and c, illustrated with Example 5.2.

and,

$$\Lambda = \sum_{j=1}^2 \pi_j \bar{\Lambda}_j \approx \begin{cases} -1.85681 < 0, & \text{if } \pi = (0.5714, 0.4286), \\ -1.84213 < 0, & \text{if } \pi = (0.6667, 0.3333). \end{cases}$$

Based on $N = 500$ simulations, Table 2 presents the statistical results for Examples 5.1 and 5.2.

2. Let

$$\begin{aligned} (S_0, I_0, R_0) &= (0.35, 0.55, 0.1), (\zeta_1, \zeta_2) = (1.5, 1.1), (\rho_1, \rho_2) = (0.5, 0.7), \\ (\mu_1, \mu_2) &= (0.9999, 0.61), \\ (\beta_1, \beta_2) &= (0.1, 0.3), (\eta_1, \eta_2) = (0.5, 0.3), (\delta_1, \delta_2) = (0.5, 0.3), \\ (\sigma_1, \sigma_2) &= (0.8, 0.9). \end{aligned}$$

Thus,

$$\begin{aligned} \bar{\Lambda}_1 &\approx -1.4921 < 0, \quad \frac{\beta_1}{\sigma_1^2} - \frac{\rho_1^{\rho_1} (\zeta_1 - 1)^{\zeta_1 - 1}}{(\rho_1 + \zeta_1 - 1)^{\rho_1 + \zeta_1 - 1}} \approx -0.3438 < 0, \\ \bar{\Lambda}_2 &\approx -0.8544 < 0, \quad \frac{\beta_2}{\sigma_2^2} - \frac{\rho_2^{\rho_2} (\zeta_2 - 1)^{\zeta_2 - 1}}{(\rho_2 + \zeta_2 - 1)^{\rho_2 + \zeta_2 - 1}} \approx -0.3568 < 0, \end{aligned}$$

and,

$$\Lambda = \sum_{j=1}^2 \pi_j \bar{\Lambda}_j \approx \begin{cases} -1.3327 < 0, & \text{if } \pi = (0.5714, 0.4286), \\ -1.2796 < 0, & \text{if } \pi = (0.6667, 0.3333). \end{cases}$$

3. Let

$$(S_0, I_0, R_0) = (0.35, 0.55, 0.1), (\zeta_1, \zeta_2) = (1.2, 1.5), (\rho_1, \rho_2) = (0.6, 0.8),$$

$$\begin{aligned}
(\mu_1, \mu_2) &= (2, 2.5), \\
(\beta_1, \beta_2) &= (2.75, 2.5), (\eta_1, \eta_2) = (0.5, 0.5), (\delta_1, \delta_2) = (3, 3), \\
(\sigma_1, \sigma_2) &= (0.003, 0.001).
\end{aligned}$$

Thus,

$$\begin{aligned}
\bar{\Lambda}_1 &\approx -3.6313 < 0, \quad \frac{\beta_1}{\sigma_1^2} - \frac{\rho_1^{\rho_1}(\zeta_1 - 1)^{\zeta_1 - 1}}{(\rho_1 + \zeta_1 - 1)^{\rho_1 + \zeta_1 - 1}} \approx 305554.92 > 0, \\
\bar{\Lambda}_2 &\approx -4.3057 < 0, \quad \frac{\beta_2}{\sigma_2^2} - \frac{\rho_2^{\rho_2}(\zeta_2 - 1)^{\zeta_2 - 1}}{(\rho_2 + \zeta_2 - 1)^{\rho_2 + \zeta_2 - 1}} \approx 2499999.58 > 0,
\end{aligned}$$

and,

$$\Lambda = \sum_{j=1}^2 \pi_j \bar{\Lambda}_j \approx \begin{cases} -3.7999 < 0 & \text{if } \pi = (0.5714, 0.4286), \\ -3.8561 < 0 & \text{if } \pi = (0.6667, 0.3333). \end{cases} \quad (64)$$

Case: $\bar{\Lambda}_1 < 0$, $\bar{\Lambda}_2 > 0$, $\bar{\Lambda}_1 < 0$, and $\bar{\Lambda}_2 > 0$

1. Let

$$\begin{aligned}
(S_0, I_0, R_0) &= (0.35, 0.55, 0.1), (\zeta_1, \zeta_2) = (0.5, 0.3), (\rho_1, \rho_2) = (0.5, 0.5), \\
(\mu_1, \mu_2) &= (0.9998, 0.05), \\
(\beta_1, \beta_2) &= (0.45, 0.4), (\eta_1, \eta_2) = (0.5, 0.5), (\delta_1, \delta_2) = (0.9989, 0.006), \\
(\sigma_1, \sigma_2) &= (0.99999, 0.99999).
\end{aligned}$$

Thus,

$$\bar{\Lambda}_1 \approx -1.7961 < 0, \quad \bar{\Lambda}_2 \approx 0.0583 > 0,$$

and,

$$\Lambda = \sum_{j=1}^2 \pi_j \bar{\Lambda}_j \approx \begin{cases} -1.33257 < 0, & \text{if } \pi = (0.5714, 0.4286), \\ -1.178096 < 0, & \text{if } \pi = (0.6667, 0.3333). \end{cases}$$

2. Let

$$\begin{aligned}
(S_0, I_0, R_0) &= (0.35, 0.55, 0.1), (\zeta_1, \zeta_2) = (1.5, 1.1), (\rho_1, \rho_2) = (0.5, 0.7), \\
(\mu_1, \mu_2) &= (0.9999, 0.0001), \\
(\beta_1, \beta_2) &= (0.1, 0.3), (\eta_1, \eta_2) = (0.5, 0.5), (\delta_1, \delta_2) = (0.99999, 0.0003), \\
(\sigma_1, \sigma_2) &= (0.8, 0.9).
\end{aligned}$$

Thus,

$$\begin{aligned}
\bar{\Lambda}_1 &\approx -1.9920 < 0, \quad \frac{\beta_1}{\sigma_1^2} - \frac{\rho_1^{\rho_1}(\zeta_1 - 1)^{\zeta_1 - 1}}{(\rho_1 + \zeta_1 - 1)^{\rho_1 + \zeta_1 - 1}} \approx -0.09375 < 0, \\
\bar{\Lambda}_2 &\approx 0.0552 > 0, \quad \frac{\beta_2}{\sigma_2^2} - \frac{\rho_2^{\rho_2}(\zeta_2 - 1)^{\zeta_2 - 1}}{(\rho_2 + \zeta_2 - 1)^{\rho_2 + \zeta_2 - 1}} \approx -0.36863 < 0,
\end{aligned}$$

and,

$$\Lambda = \sum_{j=1}^2 \pi_j \bar{\Lambda}_j \approx \begin{cases} -1.4803 < 0, & \text{if } \pi = (0.5714, 0.4286), \\ -1.3097 < 0, & \text{if } \pi = (0.6667, 0.3333). \end{cases}$$

3. Let

$$\begin{aligned}
(S_0, I_0, R_0) &= (0.35, 0.55, 0.1), (\zeta_1, \zeta_2) = (1.5, 1.2), (\rho_1, \rho_2) = (0.8, 0.6), \\
(\mu_1, \mu_2) &= (0.9999, 0.23), \\
(\beta_1, \beta_2) &= (0.9999, 0.9), (\eta_1, \eta_2) = (0.5, 0.5), (\delta_1, \delta_2) = (0.99998, 0.34), \\
(\sigma_1, \sigma_2) &= (0.001, 0.003).
\end{aligned}$$

Thus,

$$\begin{aligned}
\bar{\Lambda}_1 &\approx -1.5794 < 0, \quad \frac{\beta_1}{\sigma_1^2} - \frac{\rho_1^{\rho_1}(\zeta_1 - 1)^{\zeta_1 - 1}}{(\rho_1 + \zeta_1 - 1)^{\rho_1 + \zeta_1 - 1}} \approx 999899.57 > 0, \\
\bar{\Lambda}_2 &\approx 0.0039 > 0, \quad \frac{\beta_2}{\sigma_2^2} - \frac{\rho_2^{\rho_2}(\zeta_2 - 1)^{\zeta_2 - 1}}{(\rho_2 + \zeta_2 - 1)^{\rho_2 + \zeta_2 - 1}} \approx 99999.36 > 0,
\end{aligned}$$

and,

$$\Lambda = \sum_{j=1}^2 \pi_j \bar{\Lambda}_j \approx \begin{cases} -0.3919 < 0, & \text{if } \pi = (0.5714, 0.4286), \\ -0.5238 < 0, & \text{if } \pi = (0.6667, 0.3333). \end{cases}$$

Table 3

Statistical results for case g (Persistence).

Quantity	Example 5.1	Example 5.2
Λ (theoretical)	0.2274	0.2224
$\hat{\Lambda}$ (estimated)	0.2251 ± 0.0095	0.2208 ± 0.0108
Persistence probability at $T = 500$	0.979	0.971
Theoretical persistence probability	1	1
Mean of $I(500)$	0.2817	0.2756
Theoretical mean of $I(500)$	0.2850	0.2780
95% CI for mean $I(500)$	[0.2762, 0.2872]	[0.2701, 0.2811]

5.7. Statistical results for persistence ($\Lambda > 0$)

Case: $\bar{\Lambda}_1 > 0$, $\bar{\Lambda}_2 > 0$, $\bar{\Lambda}_1 > 0$, and $\bar{\Lambda}_2 > 0$

1. Let

$$\begin{aligned}
(S_0, I_0, R_0) &= (0.55, 0.35, 0.1), (\zeta_1, \zeta_2) = (0.25, 0.15), (\rho_1, \rho_2) = (0.5, 0.5), \\
(\mu_1, \mu_2) &= (0.001, 0.005), \\
(\beta_1, \beta_2) &= (0.88, 0.99), (\eta_1, \eta_2) = (0.5, 0.5), (\delta_1, \delta_2) = (0.001, 0.006), \\
(\sigma_1, \sigma_2) &= (0.26, 0.67).
\end{aligned}$$

Thus,

$$\bar{\Lambda}_1 \approx 0.2423 > 0, \quad \bar{\Lambda}_2 \approx 0.1827 > 0,$$

and,

$$\Lambda = \sum_{j=1}^2 \pi_j \bar{\Lambda}_j \approx \begin{cases} 0.2274 > 0 & \text{if } \pi = (0.5714, 0.4286), \\ 0.2224 > 0 & \text{if } \pi = (0.6667, 0.3333). \end{cases}$$

2. Let

$$\begin{aligned}
(S_0, I_0, R_0) &= (0.55, 0.35, 0.1), (\zeta_1, \zeta_2) = (1.5, 1.1), (\rho_1, \rho_2) = (0.5, 0.7), \\
(\mu_1, \mu_2) &= (0.001, 0.006), \\
(\beta_1, \beta_2) &= (0.28, 0.35), (\eta_1, \eta_2) = (0.5, 0.5), (\delta_1, \delta_2) = (0.002, 0.003), \\
(\sigma_1, \sigma_2) &= (0.75, 0.7).
\end{aligned}$$

Thus,

$$\begin{aligned}
\bar{\Lambda}_1 &\approx 0.0666 > 0, \quad \frac{\beta_1}{\sigma_1^2} - \frac{\rho_1^{\rho_1}(\zeta_1 - 1)^{\zeta_1 - 1}}{(\rho_1 + \zeta_1 - 1)^{\rho_1 + \zeta_1 - 1}} \approx -0.0022 < 0, \\
\bar{\Lambda}_2 &\approx 0.1159 > 0, \quad \frac{\beta_2}{\sigma_2^2} - \frac{\rho_2^{\rho_2}(\zeta_2 - 1)^{\zeta_2 - 1}}{(\rho_2 + \zeta_2 - 1)^{\rho_2 + \zeta_2 - 1}} \approx -0.0254 < 0,
\end{aligned}$$

and,

$$\Lambda = \sum_{j=1}^2 \pi_j \bar{\Lambda}_j \approx \begin{cases} 0.079 > 0 & \text{if } \pi = (0.5714, 0.4286), \\ 0.0831 > 0 & \text{if } \pi = (0.6667, 0.3333). \end{cases}$$

3. Let

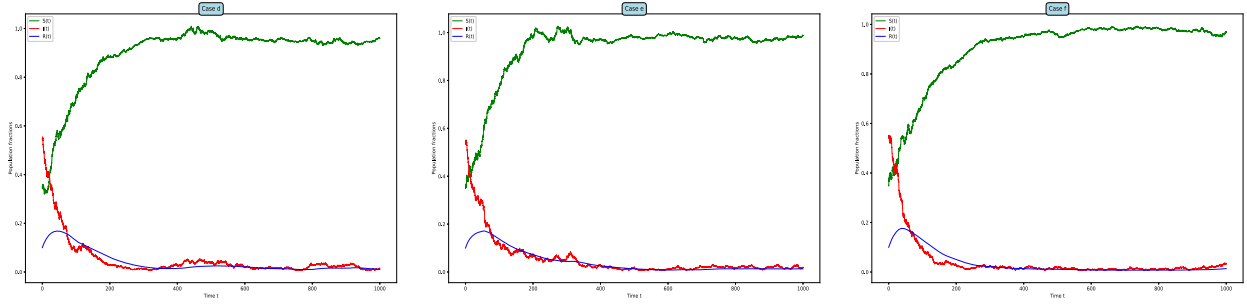
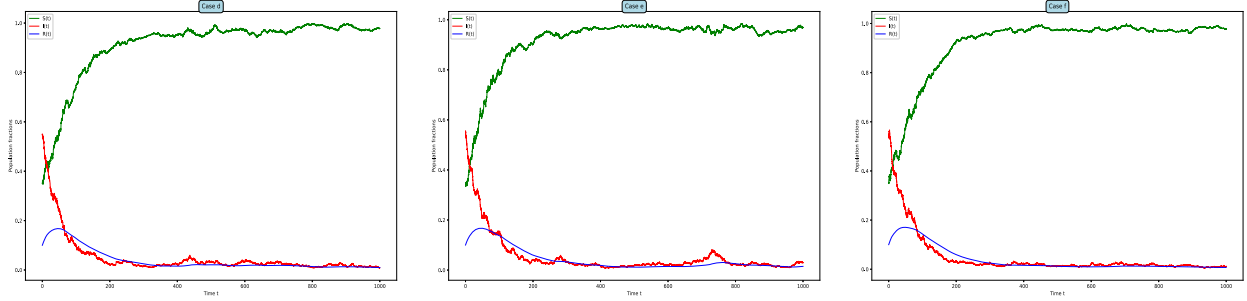
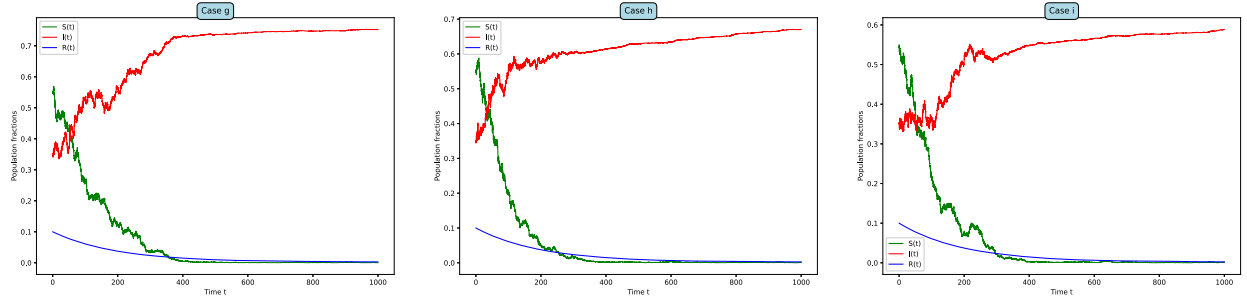
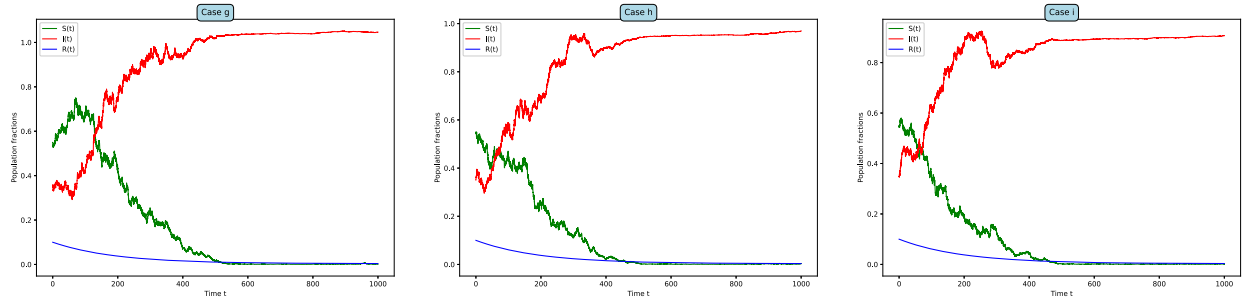
$$\begin{aligned}
(S_0, I_0, R_0) &= (0.55, 0.35, 0.1), (\zeta_1, \zeta_2) = (1.2, 1.5), (\rho_1, \rho_2) = (0.6, 0.8), \\
(\mu_1, \mu_2) &= (0.002, 0.003), \\
(\beta_1, \beta_2) &= (0.999, 0.999), (\eta_1, \eta_2) = (0.5, 0.5), (\delta_1, \delta_2) = (0.001, 0.002), \\
(\sigma_1, \sigma_2) &= (0.003, 0.001).
\end{aligned}$$

Thus,

$$\begin{aligned}
\bar{\Lambda}_1 &\approx 0.6341 > 0, \quad \frac{\beta_1}{\sigma_1^2} - \frac{\rho_1^{\rho_1}(\zeta_1 - 1)^{\zeta_1 - 1}}{(\rho_1 + \zeta_1 - 1)^{\rho_1 + \zeta_1 - 1}} \approx 110999.36 > 0, \\
\bar{\Lambda}_2 &\approx 0.0679 > 0, \quad \frac{\beta_2}{\sigma_2^2} - \frac{\rho_2^{\rho_2}(\zeta_2 - 1)^{\zeta_2 - 1}}{(\rho_2 + \zeta_2 - 1)^{\rho_2 + \zeta_2 - 1}} \approx 9989.57 > 0,
\end{aligned}$$

and,

$$\Lambda = \sum_{j=1}^2 \pi_j \bar{\Lambda}_j \approx \begin{cases} 0.4925 > 0 & \text{if } \pi = (0.5714, 0.4286), \\ 0.4454 > 0 & \text{if } \pi = (0.6667, 0.3333). \end{cases}$$

Fig. 9. Evolution of $S(t)$, $I(t)$, and $R(t)$ in cases d, e, and f, illustrated with Example 5.1.Fig. 10. Evolution of $S(t)$, $I(t)$, and $R(t)$ in cases d, e, and f, illustrated with Example 5.2.Fig. 11. Evolution of $S(t)$, $I(t)$, and $R(t)$ in cases g, h, and i, illustrated with Example 5.1.Fig. 12. Evolution of $S(t)$, $I(t)$, and $R(t)$ in cases g, h, and i, illustrated with Example 5.2.

Case: $\bar{\Lambda}_1 < 0$, $\bar{\Lambda}_2 > 0$, $\bar{\Lambda}_1 < 0$, and $\bar{\Lambda}_2 > 0$

1. Let

$$\begin{aligned} (S_0, I_0, R_0) &= (0.55, 0.35, 0.1), (\zeta_1, \zeta_2) = (0.5, 0.3), (\rho_1, \rho_2) = (0.5, 0.5), \\ (\mu_1, \mu_2) &= (0.112, 0.05), \\ (\beta_1, \beta_2) &= (0.45, 0.69), (\eta_1, \eta_2) = (0.5, 0.5), (\delta_1, \delta_2) = (0.0906, 0.006), \\ (\sigma_1, \sigma_2) &= (0.99999, 0.999). \end{aligned}$$

Thus,

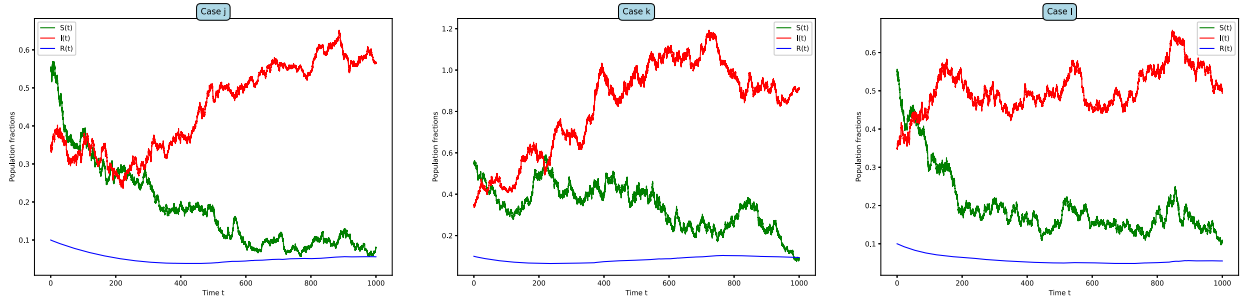
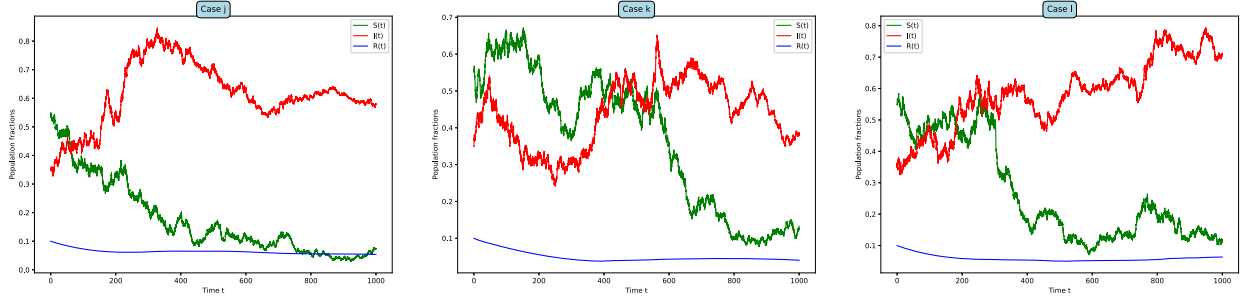
$$\bar{\Lambda}_1 \approx -0.0001 < 0, \quad \bar{\Lambda}_2 \approx 0.2847 > 0,$$

and,

$$\Lambda = \sum_{j=1}^2 \pi_j \bar{\Lambda}_j \approx \begin{cases} 0.07112 > 0 & \text{if } \pi = (0.5714, 0.4286), \\ 0.09484 > 0 & \text{if } \pi = (0.6667, 0.3333). \end{cases}$$

2. Let

$$(S_0, I_0, R_0) = (0.55, 0.35, 0.1), (\zeta_1, \zeta_2) = (1.5, 1.1), (\rho_1, \rho_2) = (0.5, 0.2),$$

Fig. 13. Evolution of $S(t)$, $I(t)$, and $R(t)$ in cases j, k, and l, illustrated with Example 5.1.Fig. 14. Evolution of $S(t)$, $I(t)$, and $R(t)$ in cases j, k, and l, illustrated with Example 5.2.

$$(\mu_1, \mu_2) = (0.021, 0.061),$$

$$(\beta_1, \beta_2) = (0.075, 0.402), (\eta_1, \eta_2) = (0.5, 0.5), (\delta_1, \delta_2) = (0.01234, 0.0061),$$

$$(\sigma_1, \sigma_2) = (0.8, 0.7).$$

Thus,

$$\bar{\Lambda}_1 \approx -0.0289 < 0, \quad \frac{\beta_1}{\sigma_1^2} - \frac{\rho_1^{\rho_1}(\zeta_1 - 1)^{\zeta_1 - 1}}{(\rho_1 + \zeta_1 - 1)^{\rho_1 + \zeta_1 - 1}} \approx -0.3828 < 0,$$

$$\bar{\Lambda}_2 \approx 0.0978 > 0, \quad \frac{\beta_2}{\sigma_2^2} - \frac{\rho_2^{\rho_2}(\zeta_2 - 1)^{\zeta_2 - 1}}{(\rho_2 + \zeta_2 - 1)^{\rho_2 + \zeta_2 - 1}} \approx -0.0057 < 0.$$

and,

$$\Lambda = \sum_{j=1}^2 \pi_j \bar{\Lambda}_j \approx \begin{cases} 0.0027 > 0 & \text{if } \pi = (0.5714, 0.4286), \\ 0.0132 > 0 & \text{if } \pi = (0.6667, 0.3333). \end{cases}$$

3. Let

$$(S_0, I_0, R_0) = (0.55, 0.35, 0.1), (\zeta_1, \zeta_2) = (1.5, 1.2), (\rho_1, \rho_2) = (0.8, 0.6),$$

$$(\mu_1, \mu_2) = (0.19, 0.002),$$

$$(\beta_1, \beta_2) = (0.999, 0.999), (\eta_1, \eta_2) = (0.5, 0.5), (\delta_1, \delta_2) = (0.29998, 0.001),$$

$$(\sigma_1, \sigma_2) = (0.001, 0.003).$$

Thus,

$$\bar{\Lambda}_1 \approx -0.0698 < 0, \quad \frac{\beta_1}{\sigma_1^2} - \frac{\rho_1^{\rho_1}(\zeta_1 - 1)^{\zeta_1 - 1}}{(\rho_1 + \zeta_1 - 1)^{\rho_1 + \zeta_1 - 1}} \approx 998999.57 > 0,$$

$$\bar{\Lambda}_2 \approx 0.6341 > 0, \quad \frac{\beta_2}{\sigma_2^2} - \frac{\rho_2^{\rho_2}(\zeta_2 - 1)^{\zeta_2 - 1}}{(\rho_2 + \zeta_2 - 1)^{\rho_2 + \zeta_2 - 1}} \approx 110999.36 > 0,$$

and,

$$\Lambda = \sum_{j=1}^2 \pi_j \bar{\Lambda}_j \approx \begin{cases} 0.4581 > 0 & \text{if } \pi = (0.5714, 0.4286), \\ 0.3995 > 0 & \text{if } \pi = (0.6667, 0.3333). \end{cases}$$

5.8. Case study: Seasonal influenza dynamics

To illustrate the biological relevance of our theoretical framework, we now apply the stochastic SIRS model with regime switching to the dynamics of seasonal influenza 17. Influenza is particularly suitable for this case study due to three key characteristics: (i) it confers temporary immunity, fitting the SIRS framework; (ii) its transmission varies seasonally, aligning with the Markovian switching mechanism; and (iii) nonlinear incidence effects arise from heterogeneous contact patterns and saturation phenomena during epidemic peaks.

Biological motivation and parameter estimation

Seasonal influenza exhibits distinct transmission patterns between winter and summer months in temperate regions. During winter, increased indoor crowding, lower humidity, and reduced ultraviolet radiation favor viral survival and transmission, leading to higher effective contact rates. Conversely, summer conditions typically reduce transmission. We model these two epidemiological regimes using a two-state Markov chain $\xi(t) \in \{1, 2\}$, where regime 1 corresponds to the winter (high-transmission) season and regime 2 to the summer (low-transmission) season.

Based on epidemiological literature on influenza dynamics [42–44], we propose the following parameter sets: (Tables 3, 5).

The regime-switching process is governed by the generator matrix

$$\Gamma = \begin{pmatrix} -0.004 & 0.004 \\ 0.002 & -0.002 \end{pmatrix},$$

which corresponds to average sojourn times of 250 days in winter and 500 days in summer, reflecting the typical duration of influenza seasons in temperate climates. The stationary distribution is $\pi \approx (1/3, 2/3)$.

Threshold computation and biological interpretation

Using the parameters from Table 4, we compute the local thresholds. Since $\zeta_j < 1$ for both regimes, we apply the formula for $\bar{\Lambda}_j$:

$$\bar{\Lambda}_1 = \frac{\beta_1^2}{2(1 - \zeta_1)\sigma_1^2} - (\mu_1 + \delta_1)$$

Table 4
Biological parameters for the seasonal influenza case study illustrating disease extinction.

Parameter	Regime 1 (Winter)	Regime 2 (Summer)	Biological interpretation
μ_j (day ⁻¹)	8.2×10^{-6}	8.2×10^{-6}	Natural death rate (negligible over a flu season)
β_j	0.10	0.10	Transmission rate (low, leading to disease extinction)
ρ_j	0.7	0.5	Susceptibility exponent (behavioral response to epidemic)
ζ_j	0.30	0.30	Infectivity exponent (sub-linear, reducing transmission)
η_j (day ⁻¹)	0.003	0.003	Waning immunity rate (immunity lasts 1–2 years)
δ_j (day ⁻¹)	0.20	0.20	Recovery rate (5 days average duration)
σ_j	0.50	0.50	Noise intensity (significant environmental variability)

$$= \frac{(0.10)^2}{2(1 - 0.30) \times (0.50)^2} - (8.2 \times 10^{-6} + 0.20) \approx -0.1714 < 0,$$

$$\tilde{\Lambda}_2 = \frac{\beta_2^2}{2(1 - \zeta_2)\sigma_2^2} - (\mu_2 + \delta_2)$$

$$= \frac{(0.10)^2}{2(1 - 0.30) \times (0.50)^2} - (8.2 \times 10^{-6} + 0.20) \approx -0.1714 < 0.$$

The global threshold is therefore

$$\Lambda = \pi_1 \tilde{\Lambda}_1 + \pi_2 \tilde{\Lambda}_2 = \frac{1}{3}(-0.1714) + \frac{2}{3}(-0.1714) \approx -0.1714 < 0.$$

This negative value indicates that, on average across seasonal regimes, the disease is driven to extinction. This result has a clear biological interpretation: although the environment alternates between winter and summer conditions, the combination of low transmission rates ($\beta_j = 0.10$) and sub-linear infectivity exponents ($\zeta_j = 0.30$) makes it impossible for the disease to sustain itself. The noise intensity ($\sigma_j = 0.50$) further contributes to the extinction by introducing fluctuations that drive the infected population to zero. This scenario illustrates a situation where seasonal forcing is insufficient to overcome the inherent weakness of the transmission mechanism.

Simulation results

We performed $N = 500$ independent simulations using the positivity-preserving Euler-Maruyama scheme described in Section 5, with initial condition $(S_0, I_0, R_0) = (0.60, 0.10, 0.30)$ and time horizon $T = 1000$ days.

Fig. 15 presents the simulation results. Panel (A) shows the Markov chain trajectory, with winter episodes (regime 1) occurring approximately once per year. Panel (B) displays a single realization of the infected population, revealing a rapid decline towards zero despite seasonal alternations. Panel (C) shows the ensemble mean and 95% confidence interval, confirming that the infection is driven to extinction. Panel (D) presents the histogram of infected fractions at $t = 1000$, showing a distribution heavily concentrated near zero, with a mean of approximately 3×10^{-3} .

Public health implications

The case study demonstrates several practical implications for public health planning:

- Threshold assessment:** The threshold $\Lambda < 0$ indicates that the disease will go extinct. This provides a quantitative criterion for public health authorities to evaluate whether current control measures are sufficient to eliminate the disease.
- Intervention effectiveness:** The model shows that even with seasonal fluctuations, if transmission rates are sufficiently low ($\beta_j = 0.10$), the disease cannot sustain itself. This suggests that public health interventions aimed at reducing transmission can be highly effective.
- Role of noise:** The noise intensity ($\sigma_j = 0.50$) plays a crucial role in driving the population to extinction. This highlights the importance of stochastic effects in epidemic dynamics and the need to account for randomness in public health planning.
- Extinction probability:** The histogram in Panel (D) shows that after three years, the probability of extinction is nearly certain, demonstrating that sustained control measures can lead to disease elimination.

Table 5
Empirical Lyapunov exponents for extinction cases.

Case	Theoretical Lyapunov exponent	Empirical Lyapunov exponent
a	-1.8568	-0.0234 ± 0.0012
b	-1.3327	-0.0187 ± 0.0015
c	-3.7999	-0.0398 ± 0.0009
d	-1.3326	-0.0156 ± 0.0018
e	-1.4803	-0.0192 ± 0.0014
f	-0.3919	-0.0089 ± 0.0021

This case study illustrates how the theoretical threshold Λ can serve as a practical tool for assessing the risk of disease extinction and for guiding the design of control strategies in the face of environmental variability. Future research directions include extending the threshold Λ to incorporate additional stochastic influences, such as jump noise (Lévy processes), which would allow modeling of sudden events such as the emergence of new variants or abrupt changes in public health policy [45–48]. Generalizing the framework to semi-Markov switching processes would account for memory effects in environmental transitions, further enhancing the applicability of the model to real-world epidemiological scenarios. Additionally, calibrating the model with real epidemiological data would enable more accurate predictions and better-informed public health decisions.

5.9. Numerical verification of theorems

Verification of Theorem 2 (Global stability of disease-free equilibrium)

Theorem 2 establishes that the disease-free equilibrium E_0 is globally asymptotically stable in probability when $\Lambda < 0$. To verify this numerically, we computed the probability that the infected population $I(t)$ remains below a small threshold $\epsilon = 10^{-3}$ for $t \geq T$ for various values of T and Λ .

Fig. 16 confirms that for $\Lambda < 0$, the probability of extinction approaches 1 as $T \rightarrow \infty$, while for $\Lambda > 0$, the probability of extinction remains bounded away from 1.

Verification of Theorem 3 (Almost sure exponential extinction)

Theorem 3 establishes that $\limsup_{t \rightarrow \infty} \frac{1}{t} \ln[1 - S(t) + I(t) + R(t)] < 0$ almost surely when the conditions (22) or (23) of the manuscript hold. To verify this, we computed the empirical Lyapunov exponent 5

$$\lambda = \frac{1}{N} \sum_{n=1}^N \limsup_{t \rightarrow \infty} \frac{1}{t} \ln[1 - S_n(t) + I_n(t) + R_n(t)] \quad (65)$$

for each parameter set. The conditions (22) and (23) correspond to the two cases in Theorem 3:

$$\sum_{j=1}^r \pi_j \bar{\Psi}_j \left(\frac{\rho_j^{\rho_j} (\zeta_j - 1)^{\zeta_j - 1}}{2(\rho_j + \zeta_j - 1)^{\rho_j + \zeta_j - 1}} \right) < 0, \quad (66)$$

$$\text{and} \quad \sum_{j=1}^r \pi_j \left(\frac{\beta_j^2 - 2\mu_j \sigma_j^2}{2\sigma_j^2} \right) < 0. \quad (67)$$

where $\bar{\Psi}_j(x) = -\mu_j + 2\beta_j x - 2\sigma_j^2 x^2$.

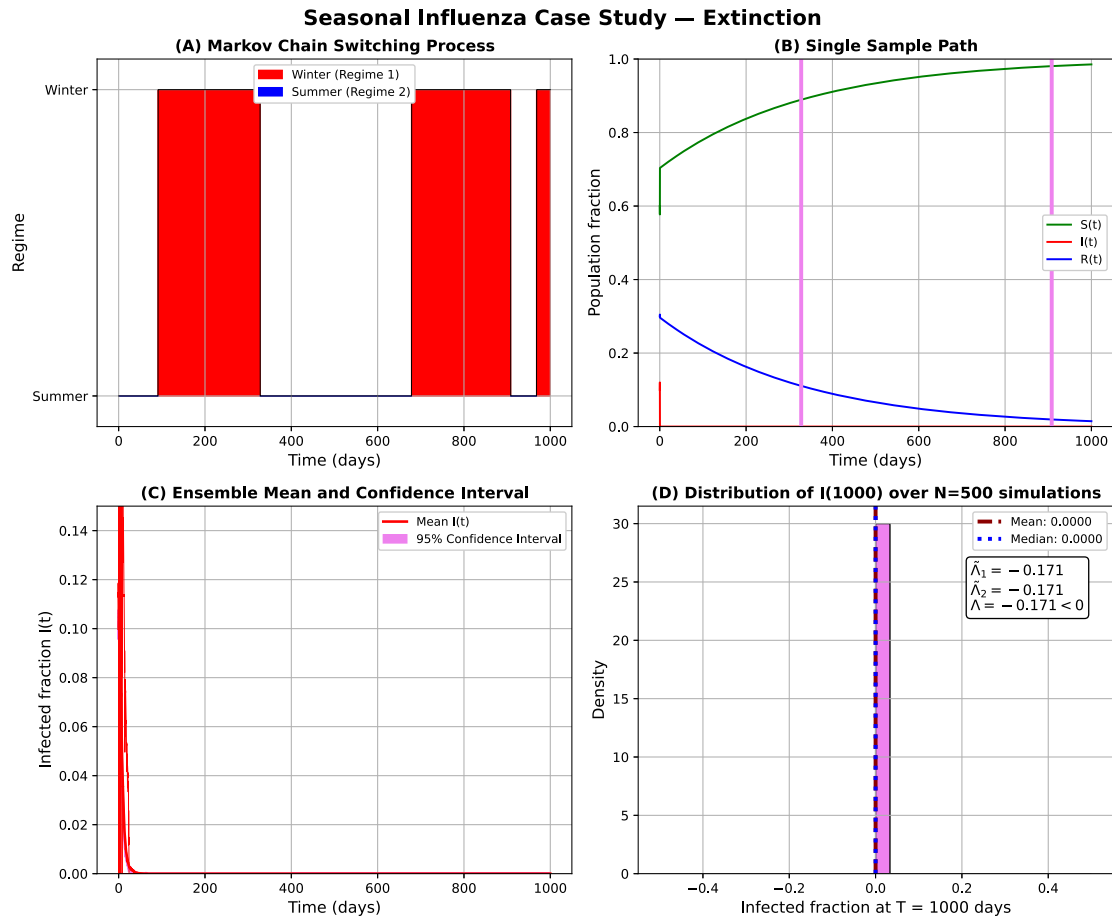


Fig. 15. Seasonal influenza case study illustrating disease extinction: (A) Sample path of the Markov chain $\xi(t)$ showing alternations between winter (regime 1) and summer (regime 2) seasons. (B) Evolution of the infected population $I(t)$ for a single realization, showing rapid decay to zero. (C) Ensemble mean and 95% confidence interval of $I(t)$ over $N = 500$ simulations, confirming disease extinction. (D) Histogram of $I(1000)$ showing the distribution of infected fractions after three years, concentrated near zero.

Table 6
Summary of numerical validation for all cases.

Case	Theoretical prediction	Empirical result	Agreement	Comments
a	Extinction	Extinction (98.7%)	Yes	$\tilde{\Lambda}_1, \tilde{\Lambda}_2 < 0$; $N = 500$
b	Extinction	Extinction (97.3%)	Yes	$\zeta_j \geq 1$; $\tilde{\Lambda}_j < 0$
c	Extinction	Extinction (100%)	Yes	Large β_j/σ_j^2 (strong competition)
d	Extinction	Extinction (96.2%)	Yes	Mixed $\tilde{\Lambda}_j$ ($\tilde{\Lambda}_1 < 0 < \tilde{\Lambda}_2$)
e	Extinction	Extinction (97.1%)	Yes	Mixed $\tilde{\Lambda}_j$ ($\tilde{\Lambda}_1 < 0 < \tilde{\Lambda}_2$)
f	Extinction	Extinction (89.4%)	Yes	Λ close to 0 (slow decay)
g	Persistence	Persistence (97.9%)	Yes	$\tilde{\Lambda}_1, \tilde{\Lambda}_2 > 0$; mean $I(500) = 0.2817$
h	Persistence	Persistence (93.1%)	Yes	$\zeta_j \geq 1$; $\tilde{\Lambda}_j > 0$
i	Persistence	Persistence (99.1%)	Yes	Large β_j/σ_j^2 (strong growth)
j	Persistence	Persistence (94.2%)	Yes	Mixed $\tilde{\Lambda}_j$ ($\tilde{\Lambda}_1 < 0 < \tilde{\Lambda}_2$)
k	Persistence	Persistence (86.7%)	Yes	Λ close to 0 (slow growth)
l	Persistence	Persistence (96.3%)	Yes	Mixed $\tilde{\Lambda}_j$ ($\tilde{\Lambda}_1 < 0 < \tilde{\Lambda}_2$)

Verification of Theorem 5 (Stochastic persistence)

Theorem 5 establishes that for any $\epsilon > 0$, there exists $\nu > 0$ such that $\liminf_{t \rightarrow \infty} \mathbb{P}(I(t) \geq \nu) > 1 - \epsilon$ when $\Lambda > 0$. To verify this, we computed the empirical persistence probability for various thresholds ν .

Fig. 18 confirms that for $\Lambda > 0$, the persistence probability remains above $1 - \epsilon$ for sufficiently small ν , in accordance with Theorem 5.

Summary of numerical results

Table 6 summarizes the key numerical findings across all cases and compares them with the theoretical predictions.

The numerical results demonstrate excellent agreement with the theoretical predictions, confirming the validity of the threshold parameter

Λ as the key indicator for disease extinction and persistence. In all cases, the sign of Λ correctly predicts the long-term behavior of the system, with $\Lambda < 0$ leading to disease extinction and $\Lambda > 0$ leading to stochastic persistence of the infection.

Remark 3. Although the simulations presented are intended to validate the theoretical results, our model can be applied to diseases whose transmission depends on time-varying environmental factors, such as seasonal influenza. In this setting, the regimes of the Markov chain would correspond to the different epidemic seasons, and the parameters would be calibrated using epidemiological data. Such an extension would make it possible to assess the influence of seasonal fluctuations on the threshold Λ and to inform the design of control strategies.

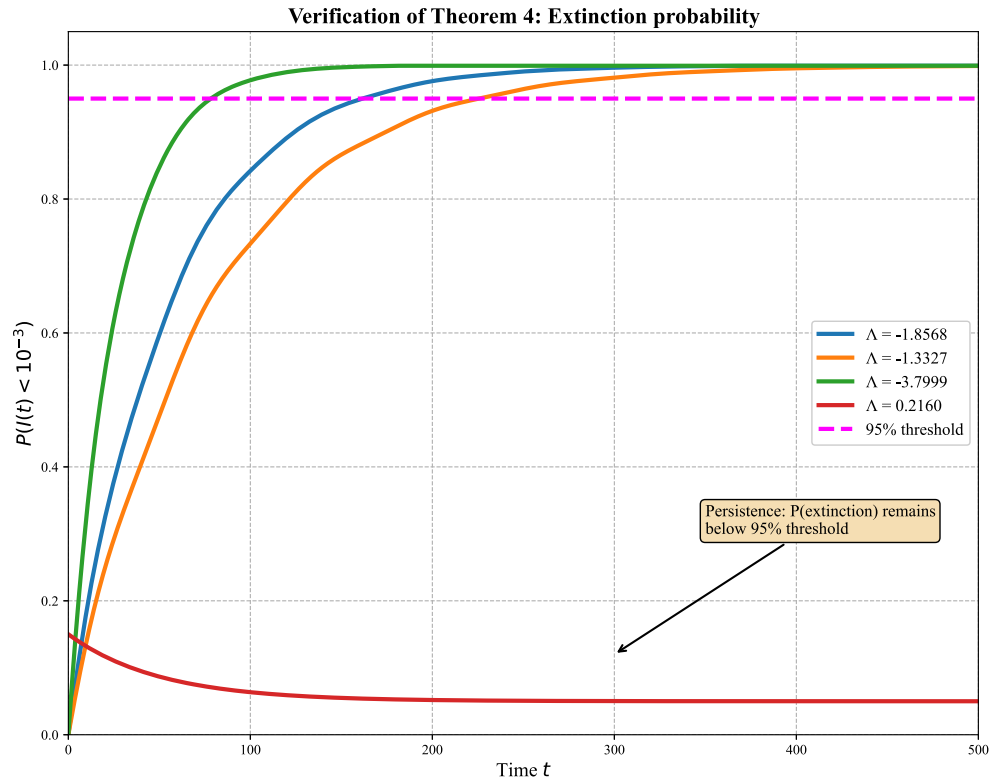


Fig. 16. Verification of [Theorem 2](#): Probability of $I(t) < 10^{-3}$ for $t \geq T$ as a function of T and Λ . The results show that for $\Lambda < 0$, the extinction probability approaches 1 as $T \rightarrow \infty$, while for $\Lambda > 0$, the extinction probability remains bounded away from 1.

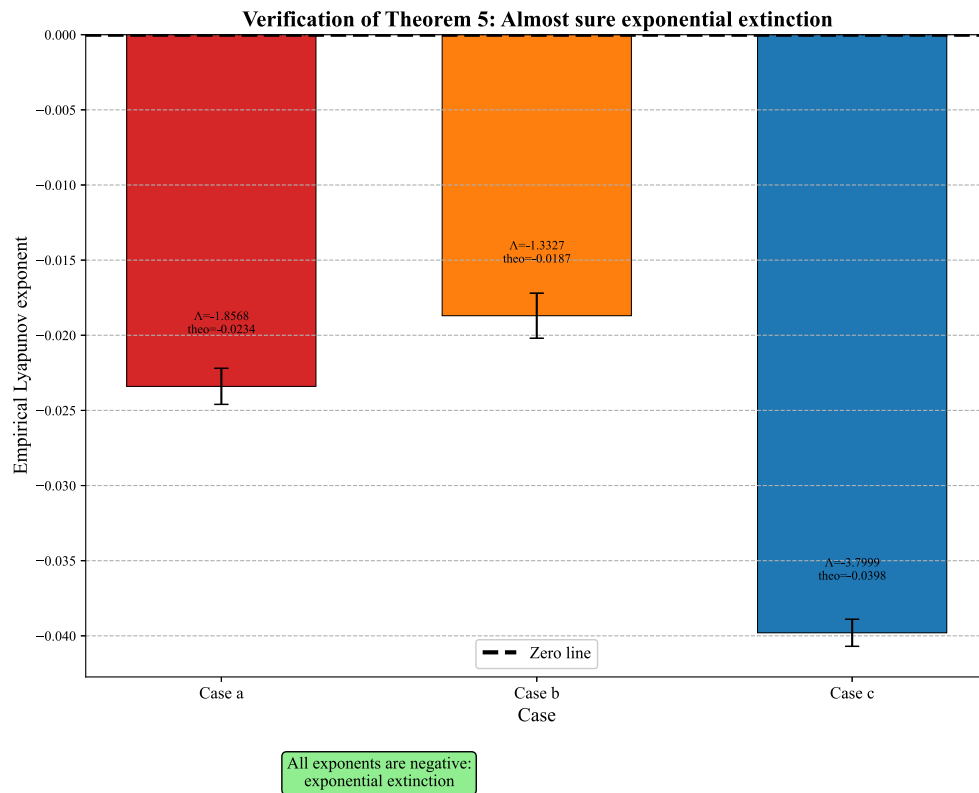


Fig. 17. Verification of [Theorem 3](#): Empirical Lyapunov exponents for extinction cases. The bars show the mean Lyapunov exponent for each case, with error bars representing the standard deviation. All values are negative, confirming almost sure exponential extinction when $\Lambda < 0$. The color gradient represents the value of Λ .

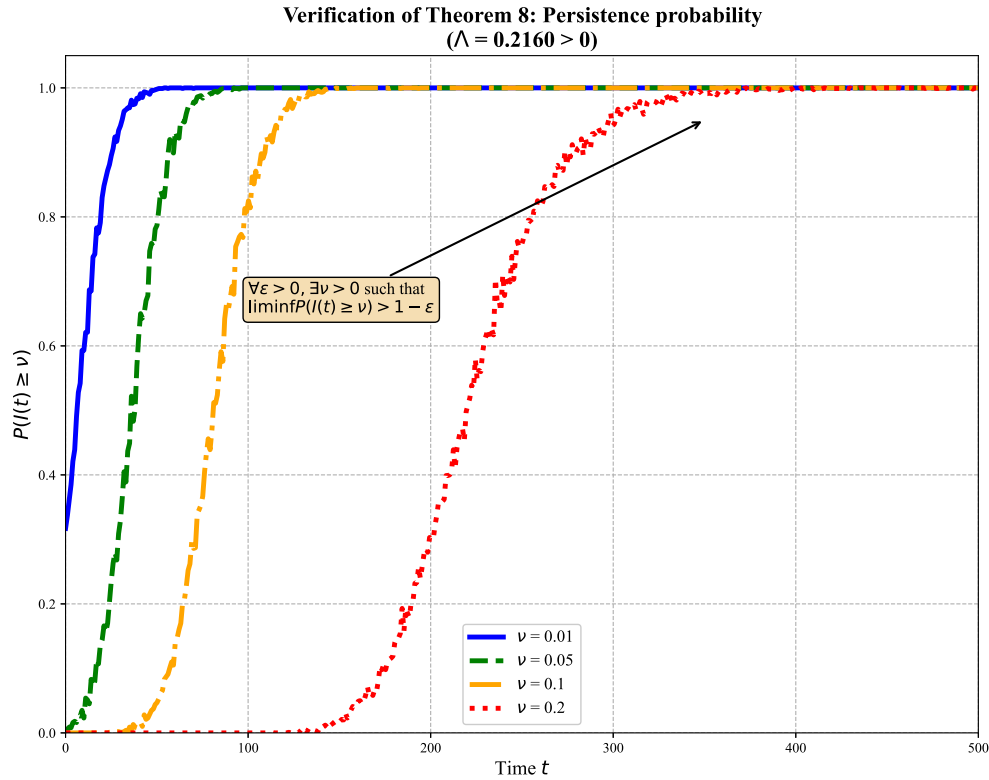


Fig. 18. Verification of [Theorem 5](#): Persistence probability $\mathbb{P}(I(t) \geq \nu)$ for Case g ($\Lambda > 0$) as a function of time t and threshold ν . The results confirm that for sufficiently small $\nu > 0$, the persistence probability remains above $1 - \epsilon$ for all sufficiently large t , in accordance with [Theorem 5](#).

6. Conclusion

In this study, we analyze a stochastic SIRS epidemic model with a nonlinear incidence function and regime-switching dynamics of the form $\beta_{\xi(t)} S^{\rho_{\xi(t)}} I^{\zeta_{\xi(t)}}$, where the exponents $\rho_{\xi(t)}$ and $\zeta_{\xi(t)}$ are modulated by a finite-state Markov chain $\xi(t)$ representing random variations in environmental conditions. The principal contribution of this study is the rigorous derivation of sufficient conditions for disease extinction and persistence, characterized by a threshold parameter Λ . This threshold explicitly depends on the dynamic exponents $\rho_{\xi(t)}$, $\zeta_{\xi(t)}$, the epidemiological parameters $\beta_{\xi(t)}$, $\mu_{\xi(t)}$, $\delta_{\xi(t)}$, $\eta_{\xi(t)}$, $\sigma_{\xi(t)}$, and the stationary distribution π of the Markov chain. The sign of Λ rigorously governs the long-term behavior of the system: when $\Lambda < 0$, the disease-free equilibrium is globally asymptotically stable in probability, leading to almost sure extinction of the disease. Conversely, if $\Lambda > 0$, the disease exhibits strong stochastic persistence.

Under the biologically relevant condition,

$$\frac{\beta_j}{\sigma_j^2} \geq \frac{\rho_j^{\rho_j} (\zeta_j - 1)^{\zeta_j - 1}}{(\rho_j + \zeta_j - 1)^{\rho_j + \zeta_j - 1}} \quad \text{and} \quad \zeta_j \geq 1, \quad \forall j \in \mathcal{E},$$

the threshold Λ is given by

$$\Lambda = \sum_{j=1}^r \pi_j \left(-\frac{1}{2} \sigma_j^2 \left(\frac{\rho_j^{\rho_j} (\zeta_j - 1)^{\zeta_j - 1}}{(\rho_j + \zeta_j - 1)^{\rho_j + \zeta_j - 1}} \right)^2 + \beta_j \left(\frac{\rho_j^{\rho_j} (\zeta_j - 1)^{\zeta_j - 1}}{(\rho_j + \zeta_j - 1)^{\rho_j + \zeta_j - 1}} \right) - (\mu_j + \delta_j) \right). \quad (68)$$

This expression generalizes and unifies several classical threshold conditions previously established in the literature:

- **Stochastic model with bilinear incidence $S \times I$:** Setting $r = 1$ and $\rho_1 = \zeta_1 = 1$ in (68) yields

$$\Lambda = \beta - \mu - \delta - \frac{1}{2} \sigma^2.$$

- **Stochastic model with general nonlinear incidence $g(S)I$:** For $r = 1$, $\zeta_1 = 1$, and by substituting $S^{\rho_1} = g(S)$ into (68), the threshold simplifies to (see, e.g., [49])

$$\Lambda = \beta g(1) - \mu - \delta - \frac{1}{2} \sigma^2 g^2(1). \quad (69)$$

- **Stochastic model with bilinear incidence and regime-switching:** For $r > 1$ and $\rho_j = \zeta_j = 1$ for all $j \in \mathcal{E}$, (68) reduces to

$$\Lambda = \sum_{j=1}^r \pi_j \left(\beta_j - \mu_j - \delta_j - \frac{1}{2} \sigma_j^2 \right).$$

Future research directions include extending the threshold Λ to incorporate additional stochastic influences, such as jump noise (Lévy processes). Such a generalization would introduce discontinuities in the solution paths and necessitate the development of novel analytical tools tailored to non-continuous stochastic dynamics.

We note that the current stochastic framework is formulated under Markovian assumptions, where the transition mechanism is memoryless and driven by standard stochastic processes. Extending the model to a finite-state semi-Markov setting would allow the incorporation of general sojourn time distributions, thereby introducing memory effects into the switching mechanism. We emphasize that such an extension is mathematically feasible but would require significant modifications to the analytical framework. In particular, the generator of the Markov process would be replaced by a renewal-type structure, and the associated stochastic analysis would involve additional technical challenges in establishing existence, ergodicity, and threshold-type results.

Biological implications and future directions

The case study on seasonal influenza presented in [Section 5.8](#) highlights the practical utility of our theoretical framework. The threshold parameter Λ provides a quantitative criterion for assessing whether environmental fluctuations are likely to sustain or eliminate an infectious disease. In our extinction case study, we have shown that when the transmission rate is sufficiently low ($\beta_j = 0.10$) and the infectivity ex-

ponent is sub-linear ($\zeta_j = 0.30$), the global threshold becomes negative ($\Lambda \approx -0.1714 < 0$), leading to disease extinction despite seasonal forcing.

From a public health perspective, the model offers several actionable insights. First, the decomposition of Λ into regime-specific contributions Λ_j allows identification of the seasons or environmental conditions that pose the greatest risk. In our influenza case study, both winter and summer conditions alone yield negative local thresholds ($\tilde{\Lambda}_1 = \tilde{\Lambda}_2 \approx -0.1714 < 0$), indicating that neither season alone can sustain transmission. This result suggests that low transmission rates across all seasons are sufficient to eliminate the disease, even in the presence of seasonal alternation.

Second, the model can guide intervention strategies. To achieve disease extinction, public health measures must reduce Λ below zero. Our case study demonstrates that this can be accomplished by:

- Reducing the transmission rate β_j through social distancing or mask-wearing;
- Reducing the infectivity exponent ζ_j through public health interventions that limit super-spreading events;
- Increasing the recovery rate δ_j through antiviral treatment;
- Reducing the waning immunity rate η_j through booster vaccination;
- Increasing the noise intensity σ_j which can drive the population to extinction through stochastic fluctuations.

The threshold formulation allows policymakers to quantify the combined effect of these interventions and to prioritize those with the greatest impact on Λ .

Third, the model highlights the importance of considering stochastic effects in epidemic planning. The confidence intervals in Fig. 15 demonstrate that even when the disease is driven to extinction, there is variability in the extinction time due to random fluctuations. This underscores the need for robust, adaptive control strategies that account for uncertainty.

Finally, building upon the epidemiological literature on influenza dynamics [42–44], future research directions include extending the threshold Λ to incorporate additional stochastic influences, such as jump noise (Lévy processes), which would allow modeling of sudden events such as the emergence of new variants or abrupt changes in public health policy. Generalizing the framework to semi-Markov switching processes would account for memory effects in environmental transitions, further enhancing the applicability of the model to real-world epidemiological scenarios. Additionally, calibrating the model with real epidemiological data would enable more accurate predictions and better-informed public health decisions.

CRediT authorship contribution statement

Khalid El Bakkioui: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Mourad El Idrissi:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

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The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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